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Effect of the vancomycin minimum inhibitory concentration on clinical outcomes in patients with methicillin-susceptible *Staphylococcus aureus* bacteraemia: a systematic review and meta-analysis

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ABSTRACT

Objective The use of the vancomycin minimum inhibitory concentration (MIC) as a prognostic predictor in patients with methicillin-susceptible *Staphylococcus aureus* (MSSA) has been debated in the last decade. We performed a systematic review and meta-analysis to investigate whether an elevated vancomycin MIC is associated with a worse prognosis for patients with MSSA bacteraemia.

Design Systematic review and meta-analysis

Data sources PubMed, EMBASE and the Cochrane Library were searched from inception to Dec 2019.

Eligibility criteria Randomized controlled trials or observational studies were considered eligible if they provided clinical outcomes of patients with MSSA bacteraemia, stratified by vancomycin MIC

Data synthesis Primary outcome was mortality. Secondary outcomes included septic thrombophlebitis, persistent bacteraemia and complicated bacteraemia. Pooled odds ratios (ORs) and 95% confidence intervals (CIs) were calculated. Subgroup analyses included susceptibility testing method.

Results Fifteen observational studies were included. Bacteraemia due to MSSA isolates with high vancomycin MICs was associated with higher mortality than isolates with low MICs (OR, 1.44; 95% CI, 1.12 to 1.84; I^2=40.3%). Additionally, significantly greater septic thrombophlebitis (OR, 3.16; 95% CI, 1.11 to 9.00; I^2=58.6%) and a trend towards more persistent bacteraemia (OR, 1.79; 95% CI, 0.97 to 3.31; I^2=0%) were observed in patients with high vancomycin MICs than in patients with low MICs. Differences in complicated bacteraemia were not significant. Similar findings were obtained in subgroup analyses using Etest. However, significant differences in outcomes were not observed between the high and low vancomycin MICs detected using broth microdilution.

Conclusions The available data suggest an association between elevated vancomycin MICs detected using Etest and adverse clinical outcomes for patients with MSSA bacteraemia. Future studies should validate these findings and explore the potential mechanisms.

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Strengths and limitations of this study

► This meta-analysis is the first to comprehensively compare the clinical outcomes of patients with methicillin-susceptible *Staphylococcus aureus* (MSSA) bacteraemia presenting high and low vancomycin minimum inhibitory concentrations (MICs).

► This meta-analysis identified an association between elevated vancomycin MICs detected using Etest and adverse clinical outcomes for patients with MSSA bacteraemia.

► Only data from observational studies were available on this topic. Selection bias likely affects the estimate of the effect size

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INTRODUCTION

Staphylococcus aureus is one of the most common causes of healthcare-associated infections. Although more publications have focused on methicillin-resistant *S. aureus* (MRSA), bacteraemia caused by methicillin-susceptible *S. aureus* (MSSA) remains a significant global healthcare burden due to its high morbidity and mortality rates [1,2].

The recognition of potential risk factors is considered vital to improve the management of patients with MSSA bacteraemia. According to previous studies, β-lactams are superior to vancomycin in terms of survival for the treatment of MSSA bacteraemia [3-5]. Therefore, the current guidelines recommend switching therapy to a β-lactam antibiotic, such as cefazolin or anti-staphylococcal penicillins (ASPs), after the identification of an MSSA infection [6]. Furthermore, our previous meta-analysis [7] systematically evaluated studies focusing on the efficacy and safety of cefazolin versus ASPs in treating MSSA bacteraemia and found that cefazolin was significantly correlated with reductions in the death rate, clinical failure, hepatotoxicity and nephrotoxicity. The results favour cefazolin for the management of MSSA bacteraemia.

The choice of the antibiotic is a well-recognized risk factor for MSSA bacteraemia, but less is known about the effects of microbiological factors. A decade ago, two studies reported an association between a high vancomycin minimum inhibitory concentration (MIC) and a worse prognosis in terms of mortality and complicated bacteraemia among patients with MSSA bacteraemia, regardless of the antibiotic therapy used [8,9]. In 2014, Kalil et al. [10] conducted a meta-analysis to identify a potential relationship between vancomycin MICs and mortality for patients with *S. aureus* bloodstream infections. Unfortunately, a subgroup analysis of MSSA bacteraemia was unable to be performed because insufficient publications were available at that time. Since then, a growing number of studies have focused on this interesting issue but have reported inconsistent results. Therefore, we conducted a systematic review and meta-analysis to assess the effect of the vancomycin MIC on clinical outcomes in patients with MSSA bacteraemia.

METHODS

Literature search

PubMed, EMBASE and the Cochrane Library were searched from inception to Dec 2019 by two

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investigators. The search terms and strategies for this meta-analysis are summarized in Table S1. References from identified publications and relevant reviews were also checked to identify additional articles. No language restriction was imposed.

Inclusion criteria and study selection

Two authors independently screened the literature, and discrepancies were resolved through discussion. Randomized controlled trials or observational studies were considered eligible if they provided clinical outcomes of patients with MSSA bacteraemia, stratified by vancomycin MIC. Broth microdilution (BMD), automated BMD (e.g. Microscan®), and Etest were considered reasonable susceptibility testing methods. The following types of studies were excluded: (a) the study population did not consist of patients with MSSA bacteraemia, (b) solely paediatric studies, (c) no outcome data were available, (d) duplicate publications, and (e) conference abstracts.

Data extraction and quality assessment

Two reviewers independently extracted the data, and discrepancies were resolved by consensus. The following information was collected: study design, countries where the study was conducted, study period, sample size, patient characteristics (age, sex, comorbidity, severity of illness, and proportions of endocarditis), vancomycin MIC testing method, cut-off value, antimicrobial therapy used, and the clinical outcomes. The Newcastle-Ottawa Scale (NOS) was used to assess the quality of observational studies [11].

Definitions and outcomes

The primary outcome was mortality. If more than one time point was available, the mortality rate reporting the largest number of deaths was used in the main analysis. The secondary outcomes included septic thrombophlebitis, persistent bacteraemia, and complicated bacteraemia, as defined by the individual. If vancomycin MIC was detected using multiple methods, the data from the reference BMD method were adopted in the main analysis. We did not define specific cut-off points for the vancomycin MIC because different criteria were used in various studies.

Statistical analysis

The odds ratios (ORs) and 95% confidence intervals (CIs) for outcomes were calculated. Heterogeneity was assessed using the I^2 statistic, and a value of > 50% was considered significant heterogeneity [12]. The fixed effects model was used when significant heterogeneity was not present;

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otherwise, the random effects model was used. Subgroup analyses were performed based on the susceptibility testing method (Etest and BMD). Sensitivity analyses were conducted for the primary outcome by removing each study individually. Additional sensitivity analyses were performed by restricting studies that were classified according to a specific criterion, such as prospective, large-sample ($n \geq 90$), high-quality (NOS score ≥ 7), adjusted mortality rate, and 30-day mortality rate (the most frequent mortality rate reported). Publication bias was estimated by constructing a funnel plot and Egger's test. All the statistical analyses in the present meta-analysis were performed using STATA software version 15.0 (StataCorp, College Station, Texas, serial NO. 301506351850). The significance threshold was set to 0.05.

RESULTS

Search results

Fifteen studies [8,9,13-25] involving 2487 participants met the inclusion criteria. The details regarding identification of relevant studies are shown in Figure 1. Two studies [17,26] contained duplicate patients and the study with smaller sample size [26] was excluded.

All studies were published within the last decade, and the number of patients per study ranged from 53 to 334. Eight studies were prospective observational studies, and the others were retrospective studies. The proportion of females ranged from 28% to 54.7%, and the mean or median age ranged from 60 to 68 years. Most studies were conducted in Europe (n=10), followed by the USA (n=3) and South Korea (n=1); the remaining study was conducted in multiple countries (Australia and New Zealand). Most of the studies used the Etest method to determine the vancomycin MIC and chose an MIC value ≥ 1.5 μg/ml as the cut-off for a high vancomycin MIC. Additional details of the characteristics are presented in Table 1. The NOS scores ranged from 5 to 8 points (Table S2).

Primary outcome: mortality

Fourteen studies reported mortality rates [8,9,13-17,19-25]. The mortality rates of MSSA bacteraemia varied among studies from 12% to 43% (Table 1). Overall, the mortality rate for patients infected with MSSA isolates with a high vancomycin MIC was significantly higher than isolates with a low MIC (OR, 1.44; 95% CI, 1.12 to 1.84; I^2=40.3%) (Figure 2). Sensitivity analyses limited to six studies providing adjusted mortality rates (OR, 1.74; 95% CI, 1.16 to 2.60; I^2=47.9%) and 12 studies

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reporting 30-day mortality rates (OR, 1.44; 95% CI, 1.09 to 1.92; I^2=28.2%) (Table 2) did not substantially alter the results of the main analysis. Similar results were obtained using the leave-one-out method (Table S3) and other prespecified sensitivity analyses (Table 3).

Secondary outcomes

Secondary outcome data were available in nine studies [8,13,15-18,21,22,24] and the definitions were showed in Table S4. Six studies reported septic thrombophlebitis [8,15-17,21,22]. An increased risk of septic thrombophlebitis was observed in patients with a high vancomycin MIC compared to patients with a low MIC (OR, 3.16; 95% CI, 1.11 to 9.00; I^2=58.6%) (Figure 3). Four studies reported persistent bacteraemia [13,17,21,22]. The meta-analysis revealed a non-significant higher risk of persistent bacteraemia in patients with a high vancomycin MIC (OR, 1.79; 95% CI, 0.97 to 3.31; I^2=0 %) (Figure 4). Seven studies reported complicated bacteraemia [8,15,16,18,21,22,24], and no significant difference was observed (OR, 1.59; 95% CI, 0.58 to 4.37; I^2=87.0%) (Figure 5).

Subgroup analysis and publication bias

In the subset of studies using Etest, patients with a high vancomycin MIC still exhibited a significantly higher mortality rate (OR, 1.62; 95% CI, 1.21 to 2.17; I^2=27.6%) and septic thrombophlebitis (OR, 4.06; 95% CI, 1.40 to 11.80; I^2=57.3%) than patients with a low MIC. However, no significant differences in outcomes were observed between the high and low vancomycin MIC values detected using broth microdilution (BMD) (Table 2). A visual inspection of the funnel plot showed symmetry (Figure S1), and the Egger test did not indicate significant publication bias (P=0.546).

DISCUSSION

To our knowledge, this systematic review and meta-analysis is the first to comprehensively compare the clinical outcomes of patients with MSSA bacteraemia presenting high and low vancomycin MICs. The main findings of the present study indicate that an elevated vancomycin MIC correlates with a higher mortality rate, regardless of the antibiotic used. The findings were relatively robust and were not significantly changed during the sensitivity analyses. In addition, patients with a high vancomycin MIC had a significantly higher risk of septic thrombophlebitis and a trend towards more persistent bacteraemia. No significant difference was observed in complicated bacteraemia.

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Based on these findings, the vancomycin MIC is potentially useful to predict adverse outcomes in patients with MSSA bacteraemia.

Several meta-analyses exploring the effect of the vancomycin MIC on the mortality rate in patients with a MRSA infection have produced positive correlations; the failure to achieve the pharmacokinetic/pharmacodynamic targets of vancomycin may be the main explanation [27-29]. However, vancomycin is not routinely used in clinical practice after the identification of an MSSA infection. In fact, the proportion of patients who received vancomycin for definitive therapy is less than one quarter in the included studies (Table 1). By analysing the subset of patients with MSSA bacteraemia treated with flucloxacillin, Holmes et al. observed a significant correlation between the 30-day mortality rate and a high vancomycin MIC [9]. Similar results were reported in the study by Cervera et al., which included patients who exclusively received cloxacillin [17], included in this meta-analysis. We hypothesise that a decreased vancomycin susceptibility might be a surrogate marker of a weak response to both vancomycin and other antibiotics (e.g., β-lactams) in the treatment of MSSA infections. Interestingly, MSSA isolates with decreased susceptibility to vancomycin also exhibit less susceptibility to the anti-staphylococcal penicillin and daptomycin [30]. This phenomenon might have contributed to special structural modifications in the cell walls of isolates with an elevated vancomycin MIC, particularly an increased cell wall thickness [31]. Moreover, reductions in autolysin activity and in the cell wall content of penicillin binding protein 4 may be involved in these changes [32].

The elevated vancomycin MIC might also be a marker for unfavourable bacterial or genetic factors that lead to adverse outcomes in patients with MSSA bacteraemia. The accessory gene regulator (*agr*) is a quorum-sensing regulator that is responsible for the expression of various virulence factors [33]. *Agr* dysfunction and *agr* type II have been correlated with an elevated vancomycin MIC, increased mortality rate, more persistent bacteraemia, and increased biofilm formation [25,34-37]. Moreover, we noticed that *agr* type III was less prevalent in MSSA isolates with a high vancomycin MIC than in isolates with a low MIC (17.2% vs. 34.0%) in the study by Lopez-Cortes et al. [21], although this difference did not reach statistical significance. In fact, the negative correlation between *agr* type III and mortality was observed in MSSA infective endocarditis [19].

Multiple clonal complexes (CCs), including CC5, CC8, CC88, and CC188, are associated with

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reduced vancomycin susceptibility in *S. aureus* bacteraemia isolates [25,34]. A recent study [38] analysed the genotypic profiles among 93 MSSA isolates reported in the study by Cervera et al. [17] and found that CC5 was significantly correlated with a higher mortality rate in the univariate analysis. Another included study [19] also revealed a correlation between MSSA isolates harbouring CC8 and increased in-hospital mortality in the univariate analysis.

Furthermore, specific resistance and virulence genes (including *blaZ, sea, clfA,* and *splA*) and the arginine catabolic mobile element locus were linked with an elevated vancomycin MIC in *S. aureus* bacteraemia [34]. MSSA isolates with genotypes *spa*-002 and *spa*-CC008 correlate with an elevated vancomycin MIC [24], and isolates with extracellular fibrinogen binding protein and serine endopeptidase are associated with a higher incidence of embolism [26].

Overall, the mechanism underlying the association between the increased vancomycin MIC and adverse outcomes in patients with MSSA bacteraemia appear to be complex. We postulate that this correlation is not attributed to a single cause, but results from the interactive effects of multiple factors. The findings of the present meta-analysis highlight the need for further studies to fill the knowledge gap about this interesting issue.

Notably, the most frequently used test to detect the vancomycin MIC in the included studies was the Etest. We identified a few studies performed susceptibility testing using Etest in parallel with manual or automated BMD methods. According to Han et al., the vancomycin MIC detected using Etest poorly correlated with the values obtained using the BMD method [20]. Song et al. reported the consistent underestimation of the MIC values from Etest among isolates with a high vancomycin MIC (2 mg/L) and overestimated among isolates with a low MIC (0.5 mg/L) using BMD [23]. As shown in the study by Sullivan et al., the vancomycin MIC values measured using automated BMD (Microscan®) are consistently higher than those the values measured using Etest in the majority of MSSA isolates [24]. Because the application of the various susceptibility testing methods may introduce bias, we conducted a subgroup analysis according to the MIC methodology. In the subgroup analysis of studies using Etest, patients with a high vancomycin MIC still had a significantly higher mortality rate than patients with a low MIC. However, no significant difference was observed in the subgroup analysis of studies using BMD. Notably, we did not include the study by Holmes et al. [9] in the subgroup analysis of studies using BMD due to the unavailability of data,

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although the authors stated that a higher mortality rate was correlated with an increased vancomycin MIC detected using BMD. Therefore, the results of the subgroup analysis of studies using BMD may be underestimated. Nevertheless, the present meta-analysis indicated that Etest might be better for predicting clinical outcomes, although the reproducibility and interobserver variation may be a concern, as noted by Falcón et al. [39] Because the BMD is still the reference method for vancomycin susceptibility testing, more studies using the BMD are need to validate the association between vancomycin MIC and clinical outcomes.

The present meta-analysis has several limitations that should be considered when interpreting the results. First, only data from observational studies were available on this topic; thus, selection bias likely affects the estimate of the effect size. Although the results of the sensitivity analysis of the adjusted mortality rate were consistent with the main analysis, other unidentified variables might influence the findings. Second, as we mentioned in the INTRODUCTION section, the choice of antibiotic is closely related to the survival of patients with MSSA bacteraemia. However, the dose and duration of antibiotic therapy and the types of β-lactams chosen were not reported in most included studies. Due to insufficient details regarding both empirical and definitive treatments, a subgroup analysis based on the antibiotic treatment was unable to be conducted. Moreover, our previous findings regarding the differences in efficacy and safety between cefazolin and ASPs have not been sufficiently considered in the included studies [7]. Third, most MSSA isolates in the included studies had a vancomycin MIC ≤ 2 mg/L determined using BMD or ≤ 1.5 mg/L determined using Etest, and the outcomes of MSSA bacteraemia caused by more resistant isolates are unclear.

In conclusion, the available data suggest a correlation between a high vancomycin MIC detected using Etest and adverse clinical outcomes for patients with MSSA bacteraemia. More well-designed studies are warranted to validate the present findings and to explore the potential mechanisms.

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Contributors NM contributed to conception and design; all authors contributed to acquisition, analysis, and interpretation of data; CS were involved in drafting or critical revision of the manuscript. All the authors approved the final version.

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isolates. ***Clin Infect Dis*** 2015;61: 1490-2.

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Table 1 Characteristics of the studies included in the meta-analysis

Study	Study design	Location	Study period	Sample size	Female %	Age, year	Assay method (HVM cut-off, mg/L)	Comorbidity & disease severity scores [a]	IE % [a]	Empirical treatment [a]	Definitive treatment [a]	Mortality %
Abelenda 2018 [13]	Ret	Spain	2014-2016	58	41.4	68	BMD (≥2.0)	NR	5.2	Consistent with guidelines 29.31%, VAN 24.13%, DAP 1.72%	VAN 22.41%, DAP 5.17 %, LIN 6.89%, CLO 39.65%	25.9
Aguado. 2011 [8]	Ret	Spain	2002-2004	99	33.3	63	Etest (≥1.5)	Mean CCI 3.4 vs 2.8	7.1	GLY 78.3% vs 60.5%; ASBL 21.7% vs 26.3%	NR	14.1
Baxi 2016 [14]	Pro	USA	2008-2013	228	NR	NR	MicroScan® (≥2.0)	NR	NR	NR	NR	15.2
Bouiller 2018 [15]	Pro	France	2009-2011	250	40.8	68	Etest (≥1.5)	NR	8.8	ASBL 57.8% vs 58.6%; VAN 9.4% vs 16.1%	Appropriate first line 96.9% vs 95.2%, optimal first line 15.6% vs 11.8%	25.6
Caston 2014 [16]	Ret	Spain	NR	53	54.7	68	MicroScan® (≥2.0)	Mean CCI 3.0 vs 2.9	5.7	ASBL 54.5% vs 40.6%; GLY 4.5% vs 19.4%; others 38.1% vs 34.3%	NR	22.6
Cervera 2014 [17]	Pro	Spain	1995-2011	93	28.0	60	Etest (≥1.5)	Median CCI 2 vs 1	100	NR	CLO 100% vs 100%	43.0
Falcon [18]	Ret	Spain	2013-2018	113	NR	NR	BMD (≥0.75 or 0.5), Etest (≥1.5)	NR	NR	NR	NR	NR
Fernandez-Hidalgo 2018 [19]	Pro	Spain	2013-2016	173	39.1	65	Etest (≥1.5)	NR	100	ASBL 100% vs 100%	NR	33.5
Han 2012 [20]	Ret	USA	2007-2009	202	NR	NR	Etest (> 1.0)	NR	NR	VAN 60.9%	β-lactams 79.2%	12.4
Holmes 2011 [9]	Pro	Australia and New Zealand	2007-2008	330	32.4	60	Etest (> 1.5), BMD (NR)	NR	10.0	NR	FLU 80.6%, VAN 19.4%	13.9
Lopez-Cortes 2015 [21]	Pro	Spain	2008-2011	135	41.5	67	Etest (≥1.5)	Median CCI 1 vs 2, median PBS 1 vs 2	8.1	NR	CLO 76.9% vs 85.7%, GLY 0 vs 8.2%, ASBL other than CLO 15.4% vs 5.1%	27.4
San-Juan 2016 [22]	Pro	Spain	2011-2014	83	41.0	60	Etest (≥1.5)	Mean CCI 4.0 vs 3.5, mean PBS 1.1 vs 1.5	2.4	ASBL 51.1% vs 52.5%, GLY 37.2% vs 40%, DAP 23.2% vs 17.5%	ASBL 100% vs 96.25%	12.0
Song 2017 [23]	Pro	South Korea	2009-2011	334	NR	NR	BMD (≥2.0), Etest (≥1.5)	NR	7.2	NR	Cephalosporin 48.1% vs 43.8%, NAF 29.6% vs 24.9%, GLY 22.2% vs 31.3%	24.9
Sullivan 2017 [24]	Ret	USA	2010-2012	252	41.3	60	MicroScan® (≥2.0), Etest (NR)	Mean CCI 5.1 vs 5.5, mean PBS 2.4 vs 2.2	NR	VAN 77% vs 62%; ASBL 17% vs 33%	NR	26.2
Viedma 2014 [25]	Ret	Spain	2010-2011	84	39.3	62	Etest (≥1.5)	NR	NR	NR	β-lactams 58.2% vs 65.5%, VAN 3.6% vs 24.1%	38.1

Abbreviations: ASBL, anti-staphylococcal β-lactams; BMD, broth microdilution; CCI, Charlson Comorbidity Index; CLO, cloxacillin; DAP, daptomycin; FLU, flucloxacillin; GLY, glycopeptides; HVM, high vancomycin minimum inhibitory concentration; IE, infective endocarditis; LIN, linezolid; LVM, low vancomycin minimum inhibitory concentration; MIC, minimum inhibitory concentration; NAF, nafcillin; NR, not reported; PBS, Pitt bacteraemia score; Pro, prospective; Ret, retrospective; VAN, vancomycin.

[a] High vancomycin MIC vs low vancomycin MIC.

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Table 2 Subgroup analyses of outcomes

Outcomes	Susceptibility testing method	No. of studies included	OR (95% CI)	P Value	I^2 %
Mortality	Etest	10	1.62 (1.21-2.17)	0.001	27.6
	BMD	5	1.12 (0.73-1.71)	0.617	45.7
Septic	Etest	5	4.06 (1.40-11.80)	0.031	57.3
thrombophlebitis	BMD	1	Not available	Not available	Not available
Persistent bacteraemia	Etest	3	1.56 (0.76-3.21)	0.224	0
	BMD	1	Not available	Not available	Not available
Complicated	Etest	5	1.92 (0.45-8.25)	0.378	90.1
bacteraemia	BMD	3	1.43 (0.58-3.51)	0.438	59.0

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Table 3 Sensitivity analyses of mortality

Term included	No. of studies included	OR (95% CI)	P Value	I^2 %
Prospective	8	1.42 (1.04-1.93)	0.026	13.0
Large-sample (n ≥ 90)	10	1.48 (1.04-2.11)	0.012	37.6
High-quality (NOS score ≥ 7)	7	1.63 (1.14-2.32)	0.007	37.5
Adjusted mortality	6	1.74 (1.16-2.60)	0.007	47.9
30-day mortality	12	1.44 (1.09-1.92)	0.011	28.2

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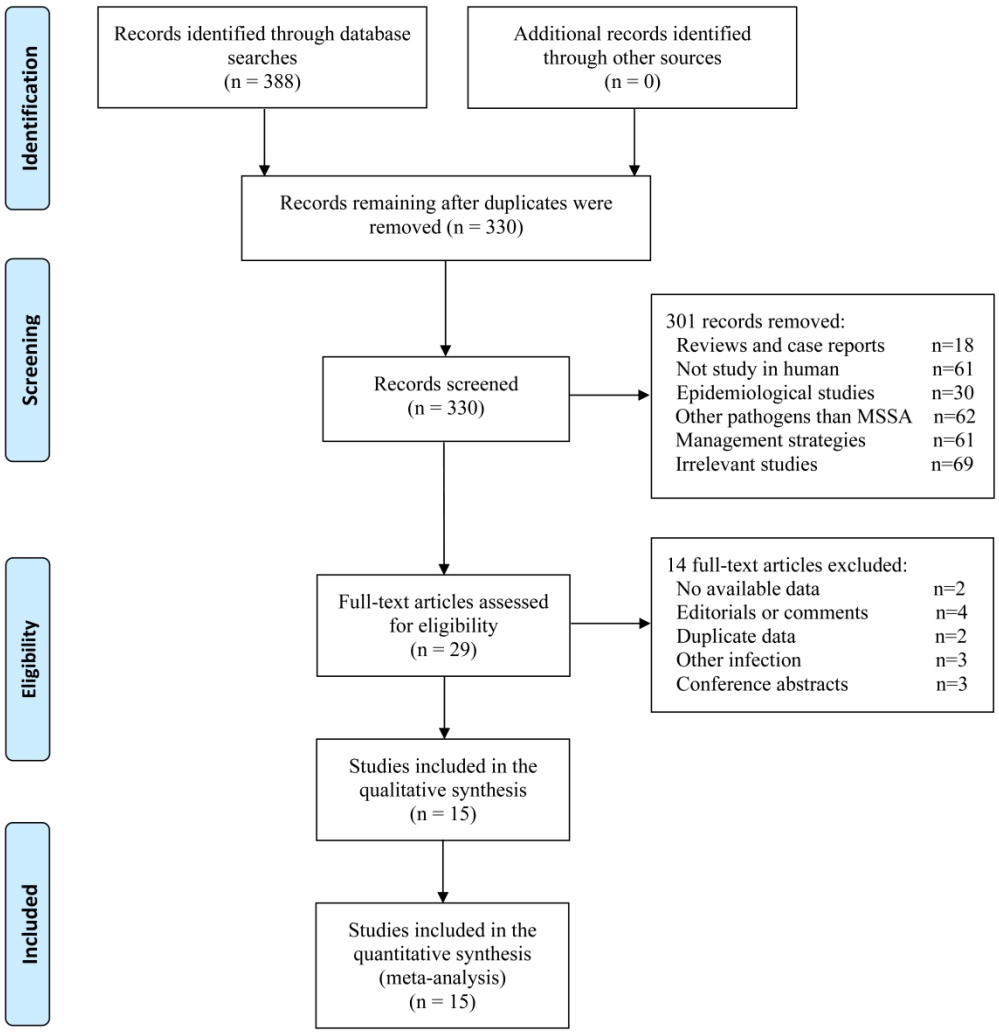


Figure 1 Flow diagram of the literature search and selection process.

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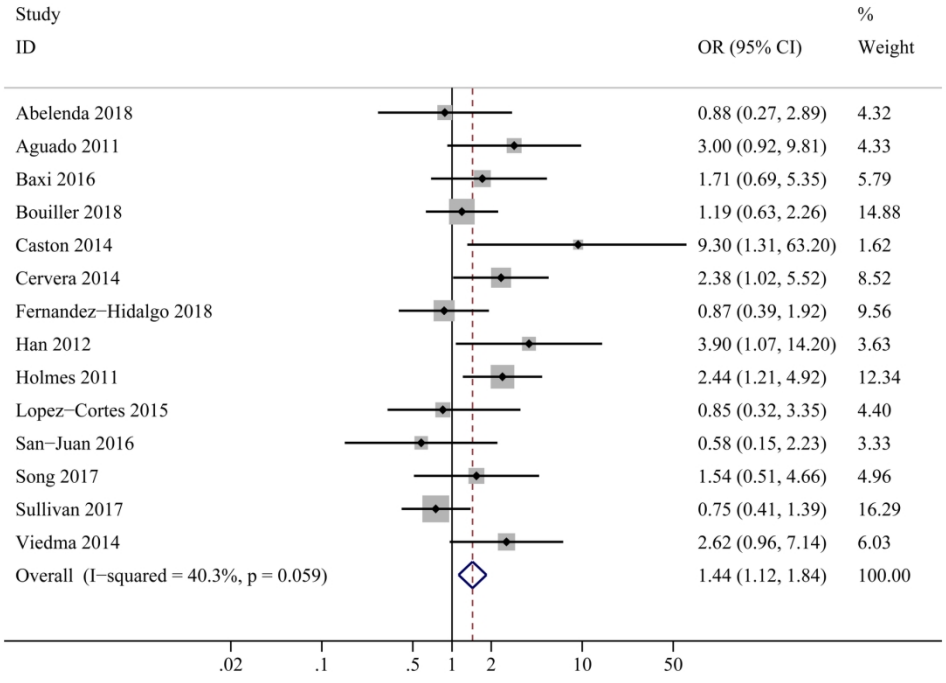


Figure 2 Forest plot of odds ratios for mortality.

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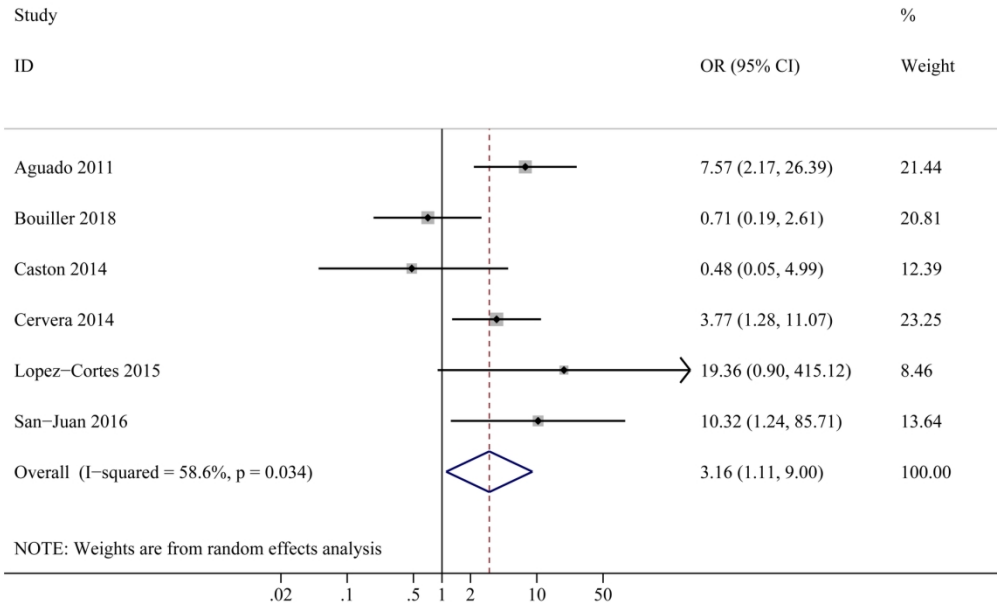


Figure 3 Forest plot of odds ratios for septic thrombophlebitis.

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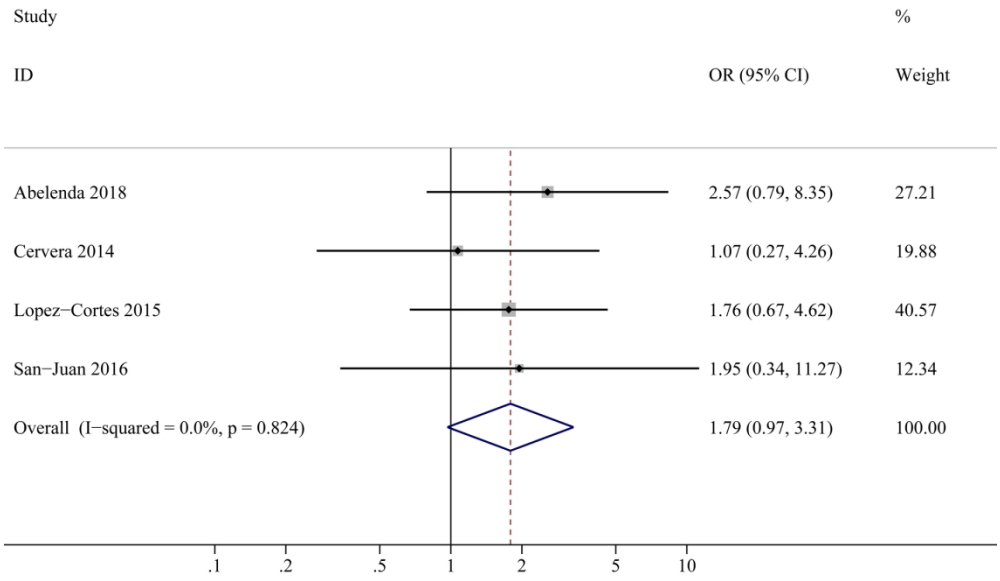


Figure 4 Forest plot of odds ratios for persistent bacteraemia.

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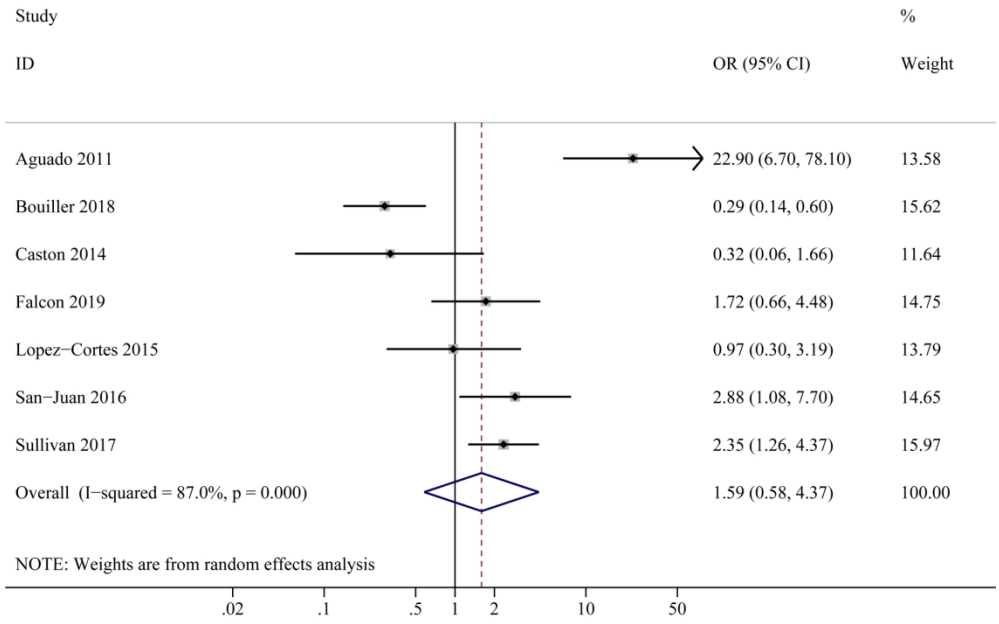


Figure 5 Forest plot of odds ratios for complicated bacteraemia.

169x118mm (300 x 300 DPI)

Table S1 Search strategy used in PubMed, EMBASE and the Cochrane Library databases

Database	PubMed		EMBASE		The Cochrane Library database	
Search number	Search query	Results	Search query	Results	Search query	Results
#1	methicillin-susceptible *Staphylococcus aureus*	3018	methicillin-susceptible *Staphylococcus aureus* [exp]	6835	methicillin-susceptible *Staphylococcus aureus*	176
#2	methicillin susceptible *Staphylococcus aureus*	5420	methicillin susceptible *Staphylococcus aureus* [exp]	6835	methicillin susceptible *Staphylococcus aureus*	238
#3	methicillin-sensitive *Staphylococcus aureus*	2300	methicillin-sensitive *Staphylococcus aureus* [exp]	6746	methicillin-sensitive *Staphylococcus aureus*	70
#4	methicillin sensitive *Staphylococcus aureus*	3920	methicillin sensitive *Staphylococcus aureus* [exp]	6746	methicillin sensitive *Staphylococcus aureus*	104
#5	MSSA	3257	MSSA	8884	MSSA	113
#6	#1 OR #2 OR #3 OR #4 OR #5	9488	#1 OR #2 OR #3 OR #4 OR #5	9786	#1 OR #2 OR #3 OR #4 OR #5	333
#7	bacteremia [mh]	45061	bacteremia [exp]	56699	bacteremia	2479
#8	bacteraemia	45061	bacteraemia [exp]	49825	bacteraemia	2479
#9	bloodstream infection	27505	bloodstream infection [exp]	15618	bloodstream infection	983
#10	#7 OR #8 OR #9	66994	#7 OR #8 OR #9	70992	#7 OR #8 OR #9	3218
#11	minimum inhibitory concentration	134806	minimum inhibitory concentration [exp]	107738	minimum inhibitory concentration	1152
#12	microbial sensitivity tests [mh]	128403	microbial sensitivity tests [mh]	9307	microbial sensitivity tests	1457
#13	MIC	39349	MIC	128645	MIC	1352
#14	#11 OR #12 OR #13	147130	#11 OR #12 OR #13	137364	#11 OR #12 OR #13	3219
#15	vancomycin [mh]	29888	vancomycin [mh]	95211	vancomycin	2008
#16	#6 AND #10 AND #14 AND #15	205	#6 AND #10 AND #14 AND #15	170	#6 AND #10 AND #14 AND #15	13

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Table S2 Assessment of the quality of the included studies

Study	Total (0-9)	Selection (0-4)	Comparability (0-2)	Outcome (0-3)
Abelenda 2018	5	3	0	2
Aguado 2011	5	2	1	2
Baxi 2016	7	4	0	3
Bouiller 2018	7	4	1	2
Caston 2014	7	3	2	2
Cervera 2014	6	3	1	2
Falcón1 2019	5	3	0	1
Fernandez-Hidalgo 2017	5	3	0	2
Han 2012	5	3	0	2
Holmes 2011	7	4	0	3
Lopez-Cortes 2015	8	4	2	2
San-Juan 2016	7	3	2	2
Song 2017	6	3	0	3
Sullivan 2017	5	3	0	2
Viedma 2014	7	3	2	2

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Table S3 Leave-one-out sensitivity analysis of mortality

Study omitted	OR (95% CI)	P Value	I^2 %
Abelenda 2018	1.47 (1.14-1.89)	0.003	43.1
Aguado 2011	1.39 (1.08-1.79)	0.01	40.7
Baxi 2016	1.42 (1.10-1.83)	0.007	44.6
Bouiller 2018	1.48 (1.14-1.94)	0.004	43.9
Caston 2014	1.39 (1.09-1.79)	0.009	33.9
Cervera 2014	1.37 (1.06-1.77)	0.016	40.8
Fernandez-Hidalgo 2018	1.52 (1.17-1.96)	0.002	40.3
Han 2012	1.38 (1.08-1.78)	0.011	38.2
Holmes 2011	1.33 (1.03-1.74)	0.032	37.8
Lopez-Cortes 2015	1.47 (1.14-1.89)	0.003	42.8
San-Juan 2016	1.48 (1.15-1.90)	0.002	40.0
Song 2017	1.43 (1.11-1.84)	0.005	44.9
Sullivan 2017	1.63 (1.25-2.13)	< 0.001	27.6
Viedma 2014	1.38 (1.07-1.78)	0.013	40.9

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Table S4 Definitions of the secondary outcomes

Study	Outcomes	Definitions of the secondary outcomes
Abelenda 2018 [13]	PB	PB was defined as positive blood cultures after more than 48 hours of appropriate antibiotic therapy.
Aguado. 2011 [8]	ST, CB	ST was defined by persistent bacteraemia at least 72 hours after initiation of active antimicrobial drugs and documented thrombi. CB was defined by one of the following events occurring after the first episode of bacteraemia: 1) development of endocarditis, septic thrombophlebitis, arthritis, spondylitis, as well as end-organ hematogenous spread of infection to other locations; 2) infection involving vascular or osteoarticular prostheses (excluding intravascular catheter) not removed within 4 days.
Bouiller 2018 [15]	ST, CB	The definition of ST was not mentioned. The definition of CB was the same as Aguado et al [8].
Caston 2014 [16]	ST, CB	The definitions of ST and CB were not mentioned.
Cervera 2014 [17]	ST, PB	The definition of ST was not mentioned. PB was defined as positive blood cultures ≥ 7 days after antimicrobial treatment.
Falcon [18]	CB	CB was defined by the following events: 1) endocarditis; 2) septic thrombophlebitis; 3) metastatic infection; 4) persistent bacteraemia for > 72 h after initiating active antimicrobial drug therapy, and 5) recurrent bacteraemia.
Lopez-Cortes 2015 [21]	ST, PB, CB	The definition of ST and CB were the same as Aguado et al [8]. PB was defined as the isolation of *S. aureus* in blood cultures obtained from peripheral veins for ≥ 3 days.
San-Juan 2016 [22]	ST, PB, CB	ST was defined by persistent bacteraemia for >72 hours after the initiation of active therapy associated to the documentation of a thrombus at the site of catheter insertion. PB was defined as positive blood cultures after more than 72 hours of appropriate antibiotic therapy. CB was defined by one of five events occurring after the incident episode of MSSA CRBSI: 1) development of endocarditis; 2) septic thrombophlebitis; 3) septic metastatic infection, including arthritis, spondylitis, infections involving vascular or osteoarticular prostheses (excluding intravascular catheter) or end-organ hematogenous seeding to other locations remote from the primary focus; 4) persistent bacteraemia for > 72 hours; or 5) persistent fever for > 4 days after extraction of the first blood culture yielding MSSA, provided that alternative causes had been reasonably excluded.
Sullivan 2017 [24]	CB	Complicated bacteraemia was defined by metastatic or embolic seeding, endovascular infection, or a duration of > 3 days.

Abbreviations: CB, complicated bacteraemia; CRBSI, catheter-related bloodstream infections; MSSA, methicillin-susceptible *Staphylococcus aureus*; PB, persistent bacteraemia; SB, septic thrombophlebitis.

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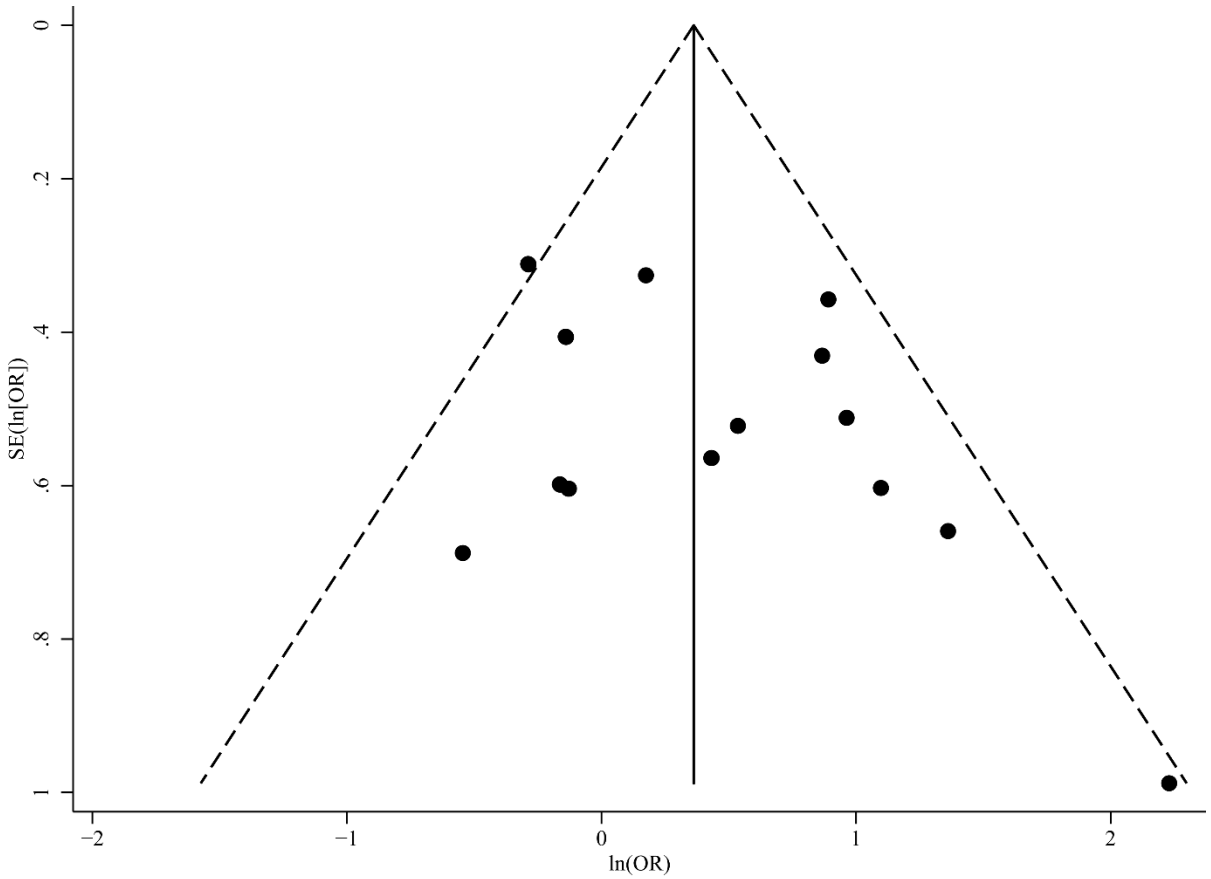


Figure S1 Funnel plot of mortality.

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	2
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	6
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	6
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	6
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	6
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	6
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	7
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	7
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	10
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	10
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	11

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Effect of the vancomycin minimum inhibitory concentration on clinical outcomes in patients with methicillin-susceptible Staphylococcus aureus bacteraemia: a systematic review and meta-analysis

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Effect of the vancomycin minimum inhibitory concentration on clinical outcomes in patients with methicillin-susceptible *Staphylococcus aureus* bacteraemia: a systematic review and meta-analysis

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ABSTRACT

Objective The use of the vancomycin minimum inhibitory concentration (MIC) as a prognostic predictor in patients with methicillin-susceptible *Staphylococcus aureus* (MSSA) has been debated in the last decade. We performed a systematic review and meta-analysis to investigate whether an elevated vancomycin MIC is associated with a worse prognosis for patients with MSSA bacteraemia.

Design Systematic review and meta-analysis

Data sources PubMed, EMBASE and the Cochrane Library were searched from inception to Dec 2019.

Eligibility criteria Randomized controlled trials or observational studies were considered eligible if they provided clinical outcomes of patients with MSSA bacteraemia, stratified by vancomycin MIC

Data synthesis Primary outcome was mortality. Secondary outcomes included septic thrombophlebitis, persistent bacteraemia and complicated bacteraemia. Pooled odds ratios (ORs) and 95% confidence intervals (CIs) were calculated. Subgroup analyses included susceptibility testing method.

Results Fifteen observational studies were included. Bacteraemia due to MSSA isolates with high vancomycin MICs was associated with higher mortality than isolates with low MICs (OR, 1.44; 95% CI, 1.12 to 1.84; I^2=40.3%). Additionally, significantly greater septic thrombophlebitis (OR, 3.16; 95% CI, 1.11 to 9.00; I^2=58.6%) and a trend towards more persistent bacteraemia (OR, 1.79; 95% CI, 0.97 to 3.31; I^2=0%) were observed in patients with high vancomycin MICs than in patients with low MICs. Differences in complicated bacteraemia were not significant. Similar findings were obtained in subgroup analyses using Etest. However, significant differences in outcomes were not observed between the high and low vancomycin MICs detected using broth microdilution.

Conclusions The available data suggest an association between elevated vancomycin MICs detected using Etest and adverse clinical outcomes for patients with MSSA bacteraemia. Future studies should validate these findings and explore the potential mechanisms.

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Strengths and limitations of this study

► This meta-analysis is the first study to comprehensively compare the clinical outcomes of patients with methicillin-susceptible *Staphylococcus aureus* bacteraemia with high and low vancomycin minimum inhibitory concentrations.

► Two authors independently performed the study selection, data extraction and quality assessment.

► Numerous sensitivity analyses were performed to examine the robustness of the findings.

► Subgroup analyses based on the susceptibility testing method were also performed.

► Due to insufficient details of empirical and definitive treatments, a subgroup analysis based on the antibiotic treatment could not be performed.

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INTRODUCTION

Staphylococcus aureus is one of the most common causes of healthcare-associated infections. Although more publications have focused on methicillin-resistant *S. aureus* (MRSA), bacteraemia caused by methicillin-susceptible *S. aureus* (MSSA) remains a significant global healthcare burden due to its high morbidity and mortality rates [1,2].

The recognition of potential risk factors is considered vital to improve the management of patients with MSSA bacteraemia. According to previous studies, β-lactams are superior to vancomycin in terms of survival for the treatment of MSSA bacteraemia [3-5]. Therefore, the current guidelines recommend switching therapy to a β-lactam antibiotic, such as cefazolin or anti-staphylococcal penicillins (ASPs), after the identification of an MSSA infection [6]. Furthermore, our previous meta-analysis [7] systematically evaluated studies focusing on the efficacy and safety of cefazolin versus ASPs in treating MSSA bacteraemia and found that cefazolin was significantly correlated with reductions in the death rate, clinical failure, hepatotoxicity and nephrotoxicity. The results favour cefazolin for the management of MSSA bacteraemia.

The choice of the antibiotic is a well-recognized risk factor for MSSA bacteraemia, but less is known about the effects of microbiological factors. A decade ago, two studies reported an association between a high vancomycin minimum inhibitory concentration (MIC) and a worse prognosis in terms of mortality and complicated bacteraemia among patients with MSSA bacteraemia, regardless of the antibiotic therapy used [8,9]. In 2014, Kalil et al. [10] conducted a meta-analysis to identify a potential relationship between vancomycin MICs and mortality for patients with *S. aureus* bloodstream infections. Unfortunately, a subgroup analysis of MSSA bacteraemia was unable to be performed because insufficient publications were available at that time. Since then, a growing number of studies have focused on this interesting issue but have reported inconsistent results. Therefore, we conducted a systematic review and meta-analysis to assess the effect of the vancomycin MIC on clinical outcomes in patients with MSSA bacteraemia.

METHODS

Literature search

PubMed, EMBASE and the Cochrane Library were searched from inception to Dec 2019 by two

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investigators. The search terms and strategies for this meta-analysis are summarized in Table S1. References from identified publications and relevant reviews were also checked to identify additional articles. No language restriction was imposed.

Inclusion criteria and study selection

Two authors independently screened the literature, and discrepancies were resolved through discussion. Randomized controlled trials or observational studies were considered eligible if they provided clinical outcomes of patients with MSSA bacteraemia, stratified by vancomycin MIC. Broth microdilution (BMD), automated BMD (e.g. Microscan®), and Etest were considered reasonable susceptibility testing methods. The following types of studies were excluded: (a) the study population did not consist of patients with MSSA bacteraemia, (b) solely paediatric studies, (c) no outcome data were available, (d) duplicate publications, and (e) conference abstracts.

Data extraction and quality assessment

Two reviewers independently extracted the data, and discrepancies were resolved by consensus. The following information was collected: study design, countries where the study was conducted, study period, sample size, patient characteristics (age, sex, comorbidity, severity of illness, and proportions of endocarditis), vancomycin MIC testing method, cut-off value, antimicrobial therapy used, and the clinical outcomes. The Risk of Bias in Nonrandomized Studies for Interventions (ROBINS-I) tool was used to assess the quality of observational studies [11].

Definitions and outcomes

The primary outcome was mortality. When data from more than one time point were available, the mortality at the latest point reporting the largest number of deaths was used in the main analysis (e.g., 30-day mortality had precedence over 14-day mortality). The secondary outcomes included septic thrombophlebitis, persistent bacteraemia, and complicated bacteraemia, as defined by the individual. If vancomycin MIC was detected using multiple methods, the data from the reference BMD method were adopted in the main analysis. We did not define specific cut-off points for the vancomycin MIC because different criteria were used in various studies.

Statistical analysis

The odds ratios (ORs) and 95% confidence intervals (CIs) for outcomes were calculated. If adjusted ORs were provided in the included studies, then they were used for data combination. Otherwise,

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unadjusted ORs were used for data combination. Heterogeneity was assessed using the I^2 statistic, and a value of > 50% was considered significant heterogeneity [12]. The fixed effects model was used when significant heterogeneity was not present; otherwise, the random effects model was used. For the outcomes using fixed effects model, we also calculated the random effects model for additional analyses to provide conservative pooling estimates. Subgroup analyses were performed based on the susceptibility testing method (Etest and BMD). Sensitivity analyses were conducted for the primary outcome by removing each study individually. Additional sensitivity analyses were performed by restricting studies that were classified according to a specific criterion, such as prospective, large-sample ($n \geq 90$), high-quality (low to moderate risk of bias in 'overall assessment' of ROBIN-I tool), unadjusted mortality rate, adjusted mortality rate, and 30-day mortality rate (the most frequent mortality rate reported). Publication bias was estimated by constructing a funnel plot and Egger's test. All the statistical analyses in the present meta-analysis were performed using STATA software version 15.0 (StataCorp, College Station, Texas, serial NO. 301506351850). The significance threshold was set to 0.05.

RESULTS

Search results

Fifteen studies [8,9,13-25] involving 2487 participants met the inclusion criteria. The details regarding identification of relevant studies are shown in Figure 1. Two studies [17,26] contained duplicate patients and the study with smaller sample size [26] was excluded.

All studies were published within the last decade, and the number of patients per study ranged from 53 to 334. Eight studies were prospective observational studies, and the others were retrospective studies. The proportion of females ranged from 28% to 54.7%, and the mean or median age ranged from 60 to 68 years. Most studies were conducted in Europe (n=10), followed by the USA (n=3) and South Korea (n=1); the remaining study was conducted in multiple countries (Australia and New Zealand). Most of the studies used the Etest method to determine the vancomycin MIC and chose an MIC value ≥ 1.5 μg/ml as the cut-off for a high vancomycin MIC. Additional details of the characteristics are presented in Table 1. Raw data for outcomes extracted from the included studies are presented in Tables S2-S5. Risk of bias assessment showed that six studies [13,15,18,22-24]

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had an overall critical risk of bias, and nine studies [8,9,14,16,17,19-21,25] had an overall low to moderate risk of bias (Table S6).

Primary outcome: mortality

Fourteen studies reported mortality rates [8,9,13-17,19-25]. The mortality rates of MSSA bacteraemia varied among studies from 12% to 43% (Table 1). Overall, the main analysis using a fixed effects model showed the mortality rate for patients infected with MSSA isolates with a high vancomycin MIC was significantly higher than isolates with a low MIC (OR, 1.44; 95% CI, 1.12 to 1.84; I^2=40.3%) (Figure 2). The additional analysis using the random effects model showed similar results (OR, 1.52; 95% CI, 1.08 to 2.12; I^2=40.3%) (Table 2). Sensitivity analyses limited to 14 studies providing unadjusted mortality rates (OR, 1.42; 95% CI, 1.13 to 1.79; I^2=36.0%), six studies providing adjusted mortality rates (OR, 1.74; 95% CI, 1.16 to 2.60; I^2=47.9%) and 12 studies reporting 30-day mortality rates (OR, 1.44; 95% CI, 1.09 to 1.92; I^2=28.2%) (Table 3) did not substantially alter the results of the main analysis. Similar results were obtained using the leave-one-out method (Table S7) and other prespecified sensitivity analyses (Table 3).

Secondary outcomes

Secondary outcome data were available in nine studies [8,13,15-18,21,22,24], and the definitions are shown in Table S8. Six studies reported septic thrombophlebitis [8,15-17,21,22]. An increased risk of septic thrombophlebitis was observed in patients with a high vancomycin MIC compared to patients with a low MIC in the pooled analysis using the random effects model (OR, 3.16; 95% CI, 1.11 to 9.00; I^2=58.6%) (Figure 3). Four studies reported persistent bacteraemia [13,17,21,22]. The meta-analysis using the fixed effects model revealed a non-significant higher risk of persistent bacteraemia in patients with a high vancomycin MIC (OR, 1.79; 95% CI, 0.97 to 3.31; I^2=0%) (Figure 4). The additional analysis using the random effects model supported the main conclusions (Table 2). Seven studies reported complicated bacteraemia [8,15,16,18,21,22,24], and no significant difference was observed in the pooled analysis using the random effects model (OR, 1.59; 95% CI, 0.58 to 4.37; I^2=87.0%) (Figure 5).

Subgroup analysis and publication bias

In the subset of studies using Etest, patients with a high vancomycin MIC still exhibited a significantly higher mortality rate (OR, 1.62; 95% CI, 1.21 to 2.17; I^2=27.6%) and septic

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thrombophlebitis (OR, 4.06; 95% CI, 1.40 to 11.80; I^2=57.3%) than patients with a low MIC. However, no significant differences in outcomes were observed between the high and low vancomycin MIC values detected using broth microdilution (BMD) (Table 2). The results of the additional analyses using the random effects model supported the main conclusions (Table 2). A visual inspection of the funnel plot showed symmetry (Figure S1), and the Egger test did not indicate significant publication bias (P=0.546).

DISCUSSION

To our knowledge, this systematic review and meta-analysis is the first to comprehensively compare the clinical outcomes of patients with MSSA bacteraemia presenting high and low vancomycin MICs. The main findings of the present study indicate that an elevated vancomycin MIC correlates with a higher mortality rate. The findings were relatively robust and were not significantly changed during the sensitivity analyses. In addition, patients with a high vancomycin MIC had a significantly higher risk of septic thrombophlebitis and a trend towards more persistent bacteraemia. No significant difference was observed in complicated bacteraemia. Based on these findings, the vancomycin MIC is potentially useful to predict adverse outcomes in patients with MSSA bacteraemia.

Several meta-analyses exploring the effect of the vancomycin MIC on the mortality rate in patients with a MRSA infection have produced positive correlations; the failure to achieve the pharmacokinetic/pharmacodynamic targets of vancomycin may be the main explanation [27-29]. However, vancomycin is not routinely used in clinical practice after the identification of an MSSA infection. In fact, the proportion of patients who received vancomycin for definitive therapy is less than one quarter in the included studies (Table 1). By analysing the subset of patients with MSSA bacteraemia treated with flucloxacillin, Holmes et al. observed a significant correlation between the 30-day mortality rate and a high vancomycin MIC [9]. Similar results were reported in the included study by Cervera et al., which included patients who exclusively received cloxacillin [17]. We hypothesise that a decreased vancomycin susceptibility might be a surrogate marker of a weak response to both vancomycin and other antibiotics (e.g., β-lactams) in the treatment of MSSA infections. Interestingly, MSSA isolates with decreased susceptibility to vancomycin also exhibit

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less susceptibility to the anti-staphylococcal penicillin and daptomycin [30]. This phenomenon might have contributed to special structural modifications in the cell walls of isolates with an elevated vancomycin MIC, particularly an increased cell wall thickness [31]. Moreover, reductions in autolysin activity and in the cell wall content of penicillin binding protein 4 may be involved in these changes [32].

The elevated vancomycin MIC might also be a marker for unfavourable bacterial or genetic factors that lead to adverse outcomes in patients with MSSA bacteraemia. The accessory gene regulator (*agr*) is a quorum-sensing regulator that is responsible for the expression of various virulence factors [33]. *Agr* dysfunction and *agr* type II have been correlated with an elevated vancomycin MIC, increased mortality rate, more persistent bacteraemia, and increased biofilm formation [25,34-37]. Moreover, we noticed that *agr* type III was less prevalent in MSSA isolates with a high vancomycin MIC than in isolates with a low MIC (17.2% vs. 34.0%) in the study by Lopez-Cortes et al. [21], although this difference did not reach statistical significance. In fact, the negative correlation between *agr* type III and mortality was observed in MSSA infective endocarditis [19].

Multiple clonal complexes (CCs), including CC5, CC8, CC88, and CC188, are associated with reduced vancomycin susceptibility in *S. aureus* bacteraemia isolates [25,34]. A recent study [38] analysed the genotypic profiles among 93 MSSA isolates reported in the study by Cervera et al. [17] and found that CC5 was significantly correlated with a higher mortality rate in the univariate analysis. Another included study [19] also revealed a correlation between MSSA isolates harbouring CC8 and increased in-hospital mortality in the univariate analysis.

Furthermore, specific resistance and virulence genes (including *blaZ, sea, clfA,* and *splA*) and the arginine catabolic mobile element locus were linked with an elevated vancomycin MIC in *S. aureus* bacteraemia [34]. MSSA isolates with genotypes *spa*-002 and *spa*-CC008 correlate with an elevated vancomycin MIC [24], and isolates with extracellular fibrinogen binding protein and serine endopeptidase are associated with a higher incidence of embolism [26].

Overall, the mechanism underlying the association between the increased vancomycin MIC and adverse outcomes in patients with MSSA bacteraemia appear to be complex. We postulate that this correlation is not attributed to a single cause, but results from the interactive effects of multiple factors. The findings of the present meta-analysis highlight the need for further studies to fill the

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knowledge gap about this interesting issue.

Notably, the most frequently used test to detect the vancomycin MIC in the included studies was the Etest. We identified a few studies performed susceptibility testing using Etest in parallel with manual or automated BMD methods. According to Han et al., the vancomycin MIC detected using Etest poorly correlated with the values obtained using the BMD method [20]. Song et al. reported the consistent underestimation of the MIC values from Etest among isolates with a high vancomycin MIC (2 mg/L) and overestimated among isolates with a low MIC (0.5 mg/L) using BMD [23]. As shown in the study by Sullivan et al., the vancomycin MIC values measured using automated BMD (Microscan®) are consistently higher than the values measured using Etest in the majority of MSSA isolates [24]. Because the application of the various susceptibility testing methods may introduce bias, we conducted a subgroup analysis according to the MIC methodology. In the subgroup analysis of studies using Etest, patients with a high vancomycin MIC still had a significantly higher mortality rate than patients with a low MIC. However, no significant difference was observed in the subgroup analysis of studies using BMD. Notably, we did not include the study by Holmes et al. [9] in the subgroup analysis of studies using BMD due to the unavailability of data, although the authors stated that a higher mortality rate was correlated with an increased vancomycin MIC detected using BMD. Therefore, the results of the subgroup analysis of studies using BMD may be underestimated. Nevertheless, the present meta-analysis indicated that Etest might be better for predicting clinical outcomes, although the reproducibility and interobserver variation may be a concern, as noted by Falcón et al. [39] Because the BMD is still the reference method for vancomycin susceptibility testing, more studies using the BMD are needed to validate the association between vancomycin MIC and clinical outcomes.

The present meta-analysis has several limitations that should be considered when interpreting the results. First, all studies included in the meta-analysis were observational studies containing many confounding factors. Therefore, selection bias and comparability bias likely affected the estimate of the effect size. For example, patients with high vancomycin MICs had numerically higher mean or median CCI score in several studies [8,17,22]. Bouiller et al. obseved more dialysis and higher proportion ultimately fatal disease in the high vancomycin MICs group [15]. Although the results of the sensitivity analysis of the adjusted mortality rate were consistent with the main analysis, other

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unidentified variables (e.g., source of infection, underlying conditions, and the calculation of severity scores) may influence the findings. Second, as we mentioned in the INTRODUCTION section, the choice of antibiotic is closely related to the survival of patients with MSSA bacteraemia. However, the dose and duration of antibiotic therapy and the types of β-lactams chosen were not reported in most included studies. Due to insufficient details regarding both empirical and definitive treatments, a subgroup analysis based on the antibiotic treatment was unable to be conducted. Moreover, our previous findings regarding the differences in efficacy and safety between cefazolin and ASPs have not been sufficiently considered in the included studies [7]. Third, most MSSA isolates in the included studies had a vancomycin MIC ≤ 2 mg/L determined using BMD or ≤ 1.5 mg/L determined using Etest, and the outcomes of MSSA bacteraemia caused by more resistant isolates are unclear.

In conclusion, the available data suggest a correlation between a high vancomycin MIC detected using Etest and adverse clinical outcomes for patients with MSSA bacteraemia. More well-designed studies are warranted to validate the present findings and to explore the potential mechanisms.

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Contributors Nengming Lin contributed to conception and design; Changcheng Shi, Jian Ye, Renjie Xu, Weizhong Jin, Shuang Xu, Fei Teng, and Nengming Lin contributed to acquisition, analysis, and interpretation of data; Changcheng Shi were involved in drafting or critical revision of the manuscript. Changcheng Shi, Jian Ye, Renjie Xu, Weizhong Jin, Shuang Xu, Fei Teng, and Nengming Lin approved the final version.

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Data availability statement All data relevant to the study are included in the article or uploaded as supplementary information

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Tables

Table 1 Characteristics of the studies included in the meta-analysis

Table 2 Subgroup analyses of mortality

Table 3 Sensitivity analyses of mortality

Figure legends

Figure 1 Flow diagram of the literature search and selection process.

Figure 2 Forest plot of odds ratios for mortality.

Figure 3 Forest plot of odds ratios for septic thrombophlebitis.

Figure 4 Forest plot of odds ratios for persistent bacteraemia.

Figure 5 Forest plot of odds ratios for complicated bacteraemia.

Supplemental data

Table S1 Search strategies used in PubMed, EMBASE and the Cochrane Library Databases

Table S2 Raw data of mortality extracted from the included studies

Table S3 Raw data of septic thrombophlebitis extracted from the included studies

Table S4 Raw data of persistent bacteraemia extracted from the included studies

Table S5 Raw data of complicated bacteraemia extracted from the included studies

Table S6 Assessment of the quality of the included studies

Table S7 Leave-one-out sensitivity analysis of mortality

Table S8 Definitions of the secondary outcomes

Figure S1 Funnel plot of mortality.

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Table 1 Characteristics of the studies included in the meta-analysis

Study	Study design	Location	Study period	Sample size	Female %	Age, year	Assay method (HVM cut-off, mg/L)	Comorbidity & disease severity scores [a]	IE % [a]	Empirical treatment [a]	Definitive treatment [a]	Mortality %	Mortality definition
Abelenda 2018 [13]	Ret	Spain	2014-2016	58	41.4	68	BMD (≥2.0)	NR	5.2	Consistent with guidelines 29.31%, VAN 24.13%, DAP 1.72%	VAN 22.41%, DAP 5.17 %, LIN 6.89%, CLO 39.65%	25.9	30-day
Aguado. 2011 [8]	Ret	Spain	2002-2004	99	33.3	63	Etest (≥1.5)	Mean CCI 3.4 vs 2.8	7.1	GLY 78.3% vs 60.5%; ASBL 21.7% vs 26.3%	NR	14.1	30-day
Baxi 2016 [14]	Pro	USA	2008-2013	230	NR	NR	MicroScan® (≥2.0)	NR	NR	NR	NR	16.1 and 11.7	90-day, 30-day
Bouiller 2018 [15]	Pro	France	2009-2011	250	40.8	68	Etest (≥1.5)	NR	8.8	ASBL 57.8% vs 58.6%; VAN 9.4% vs 16.1%	Appropriate first line 96.9% vs 95.2%, optimal first line 15.6% vs 11.8%	25.6	30-day
Caston 2014 [16]	Ret	Spain	NR	53	54.7	68	MicroScan® (≥2.0)	Mean CCI 3.0 vs 2.9	5.7	ASBL 54.5% vs 40.6%; GLY 4.5% vs 19.4%; others 38.1% vs 34.3%	NR	22.6	30-day
Cervera 2014 [17]	Pro	Spain	1995-2011	93	28.0	60	Etest (≥1.5)	Median CCI 2 vs 1	100	NR	CLO 100% vs 100%	43.0 and 39.8	One-year and in-hospital
Falcon 2019 [18]	Ret	Spain	2013-2018	113	NR	NR	BMD (≥0.75 or 0.5), Etest (≥1.5)	NR	NR	NR	NR	NR	NR
Fernandez-Hidalgo 2018 [19]	Pro	Spain	2013-2016	173	39.1	65	Etest (≥1.5)	NR	100	ASBL 100% vs 100%	NR	33.5	In-hospital
Han 2012 [20]	Ret	USA	2007-2009	202	NR	NR	Etest (> 1.0)	NR	NR	VAN 60.9%	β-lactams 79.2%	12.4	30-day
Holmes 2011 [9]	Pro	Australia and New Zealand	2007-2008	330	32.4	60	Etest (> 1.5), BMD (NR)	NR	10.0	NR	FLU 80.6%, VAN 19.4%	13.9	30-day
Lopez-Cortes 2015 [21]	Pro	Spain	2008-2011	135	41.5	67	Etest (≥1.5)	Median CCI 1 vs 2, median PBS 1 vs 2	8.1	NR	CLO 76.9% vs 85.7%, GLY 0 vs 8.2%, ASBL other than CLO 15.4% vs 5.1%	27.4 and 20.7	30-day and 14-day
San-Juan 2016 [22]	Pro	Spain	2011-2014	83	41.0	60	Etest (≥1.5)	Mean CCI 4.0 vs 3.5, mean PBS 1.1 vs 1.5	2.4	ASBL 51.1% vs 52.5%, GLY 37.2% vs 40%, DAP 23.2% vs 17.5%	ASBL 100% vs 96.25%	12.0	30-day
Song 2017 [23]	Pro	South Korea	2009-2011	334	NR	NR	BMD (≥2.0), Etest (≥1.5)	NR	7.2	NR	Cephalosporin 48.1% vs 43.8%, NAF 29.6% vs 24.9%, GLY 22.2% vs 31.3%	24.9	30-day
Sullivan 2017 [24]	Ret	USA	2010-2012	252	41.3	60	MicroScan® (≥2.0), Etest (NR)	Mean CCI 5.1 vs 5.5, mean PBS 2.4 vs 2.2	NR	VAN 77% vs 62%; ASBL 17% vs 33%	NR	26.2, 23.4 and 17.8	90-day, 60-day, and 30-day
Viedma 2014 [25]	Ret	Spain	2010-2011	84	39.3	62	Etest (≥1.5)	NR	NR	NR	β-lactams 58.2% vs 65.5%, VAN 3.6% vs 24.1%	38.1 and 20.2	In-hospital and 30-day

Abbreviations: ASBL, anti-staphylococcal β-lactams; BMD, broth microdilution; CCI, Charlson Comorbidity Index; CLO, cloxacillin; DAP, daptomycin; FLU, flucloxacillin; GLY, glycopeptides; HVM, high vancomycin minimum inhibitory concentration; IE, infective endocarditis; LIN, linezolid; LVM, low vancomycin minimum inhibitory concentration; MIC, minimum inhibitory concentration; NAF, nafcillin; NR, not reported; PBS, Pitt bacteraemia score; Pro, prospective; Ret, retrospective; VAN, vancomycin.

[a] High vancomycin MIC vs low vancomycin MIC.

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Table 2 Subgroup analyses of outcomes

Outcomes	Susceptibility testing method	No. of studies	Main analyses				Additional analyses using the random effects model	
			OR (95% CI)	P Value	I^2 %	Effects model	OR (95% CI)	P Value
Mortality	Etest	10	1.62 (1.21-2.17)	0.001	27.6	Fixed	1.63 (1.15, 2.32)	0.006
	BMD	5	1.12 (0.73-1.71)	0.617	45.7	Fixed	1.32 (0.69, 2.52)	0.394
	Overall	14	1.44 (1.12-1.84)	0.004	40.3	Fixed	1.52 (1.08-2.12)	0.015
Septic	Etest	5	4.06 (1.40-11.8)	0.031	57.3	Random	Not available	Not available
thrombophlebitis	BMD	1	Not available	Not available	Not available	Not available	Not available	Not available
	Overall	6	3.16 (1.11-9.00)	0.031	58.6	Random	Not available	Not available
Persistent	Etest	3	1.56 (0.76-3.21)	0.224	0	Fixed	1.56 (0.76-3.21)	0.224
bacteraemia	BMD	1	Not available	Not available	Not available	Not available	Not available	Not available
	Overall	4	1.79 (0.97-3.31)	0.064	0	Fixed	1.79 (0.97-3.31)	0.064
Complicated	Etest	5	1.92 (0.45-8.25)	0.378	90.1	Random	Not available	Not available
bacteraemia	BMD	3	1.43 (0.58-3.51)	0.438	59.0	Random	Not available	Not available
	Overall	7	1.59 (0.58-4.37)	0.365	87.0	Random	Not available	Not available

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Table 3 Sensitivity analyses of mortality

Term included	No. of studies included	OR (95% CI)	P Value	I^2 %
Prospective	8	1.42 (1.04-1.93)	0.026	13.0
Large-sample (n ≥ 90)	10	1.48 (1.04-2.11)	0.012	37.6
Low to moderate overall risk of bias	9	1.99 (1.43-2.76)	<0.001	27.0
Adjusted mortality	6	1.74 (1.16-2.60)	0.007	47.9
Unadjusted mortality	14	1.42 (1.13-1.79)	0.003	36.0
30-day mortality	12	1.44 (1.09-1.92)	0.011	28.2

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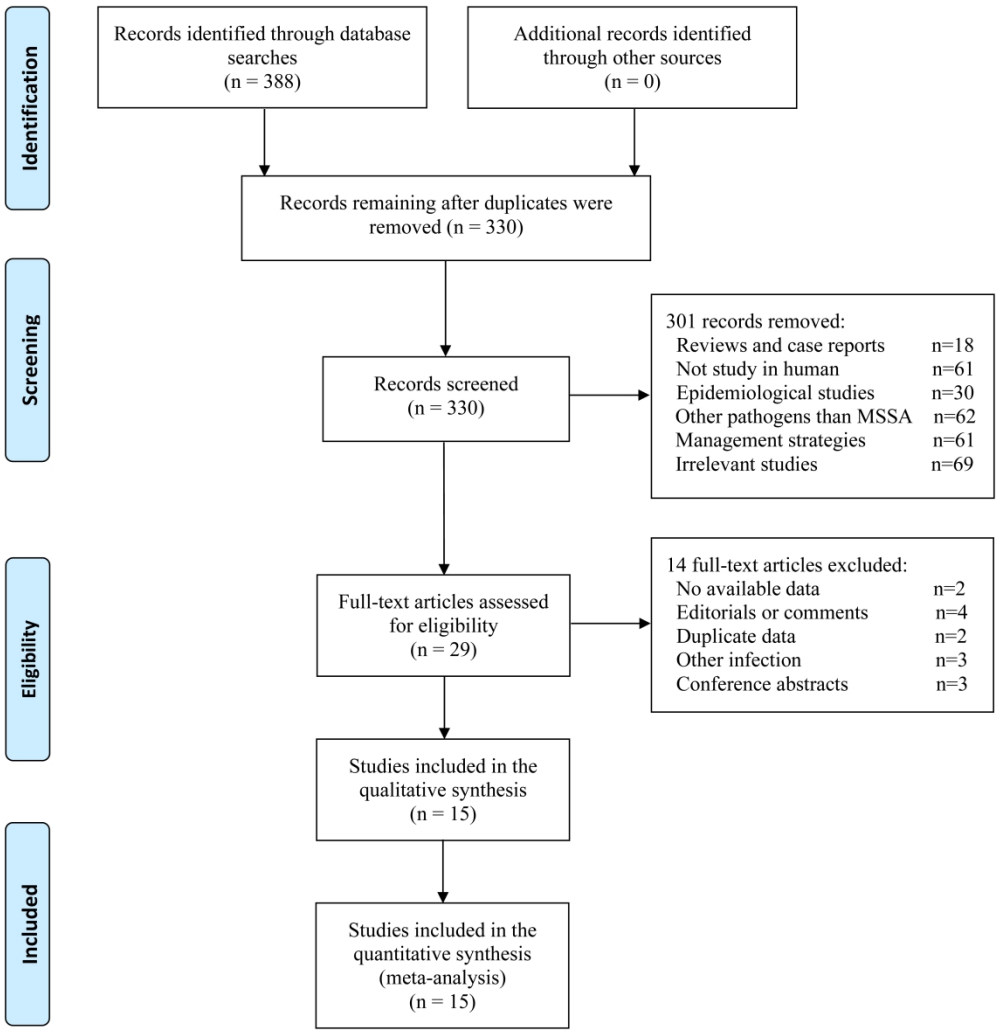


Figure 1 Flow diagram of the literature search and selection process.

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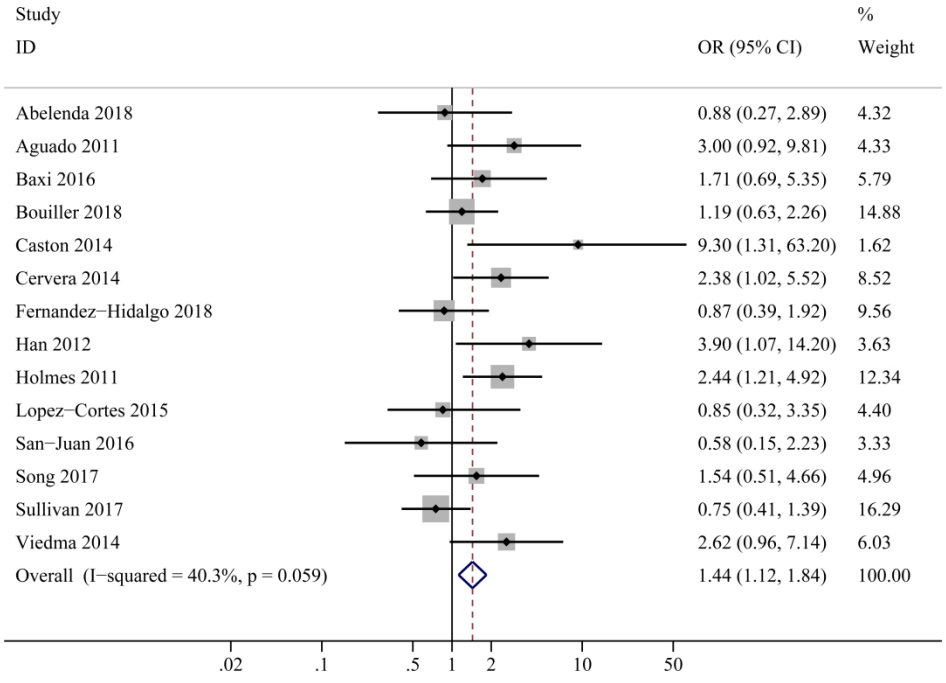


Figure 2 Forest plot of odds ratios for mortality.

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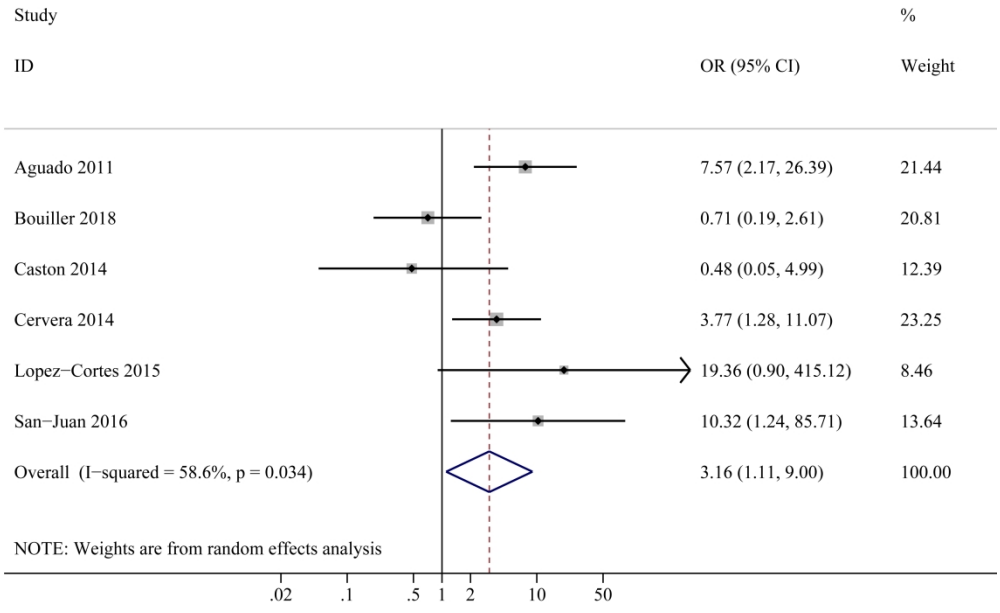


Figure 3 Forest plot of odds ratios for septic thrombophlebitis.

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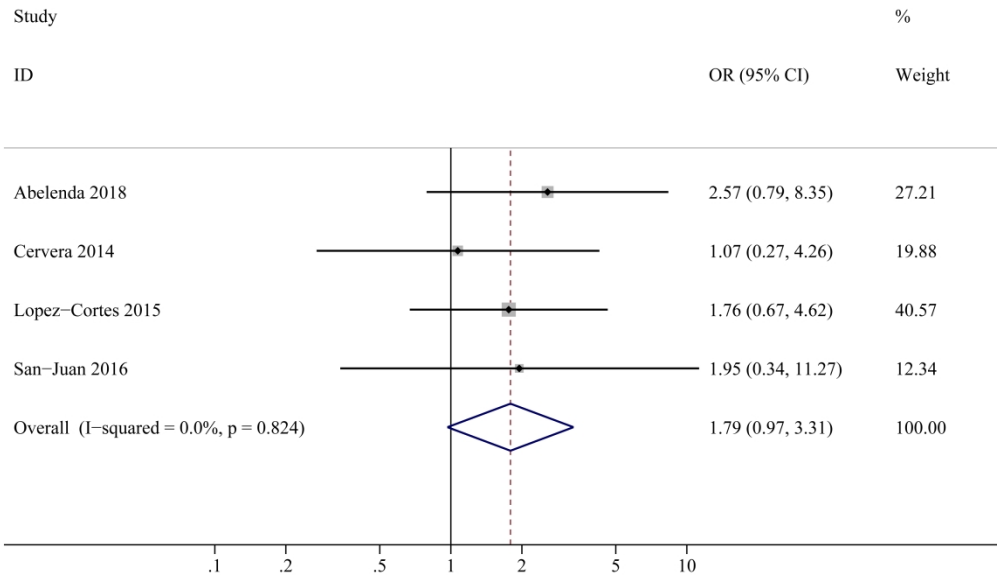


Figure 4 Forest plot of odds ratios for persistent bacteraemia.

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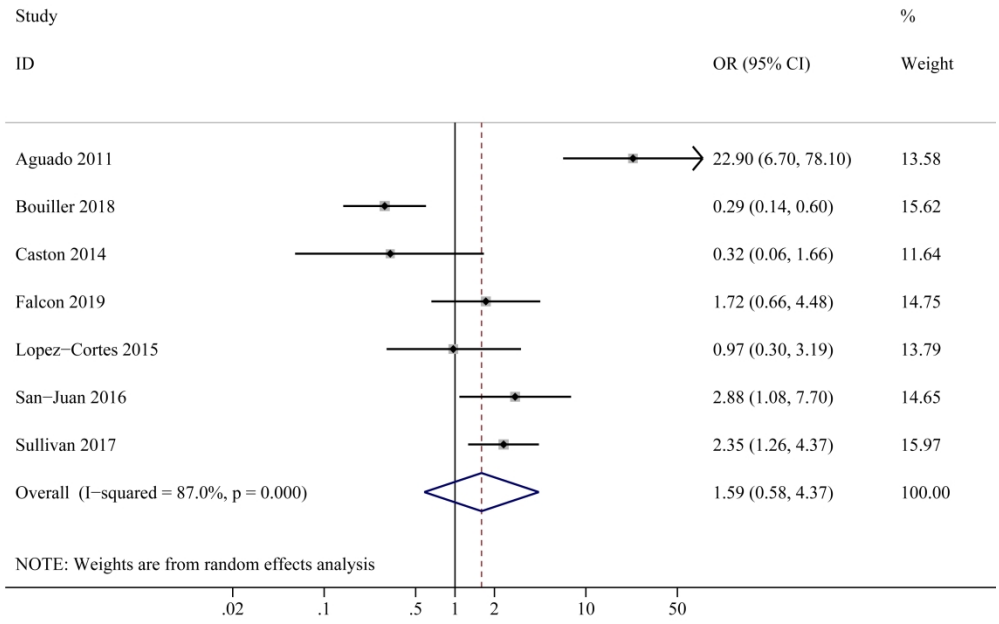


Figure 5 Forest plot of odds ratios for complicated bacteraemia.

169x118mm (600 x 600 DPI)

Table S1 Search strategy used in PubMed, EMBASE and the Cochrane Library databases

Database	PubMed		EMBASE		The Cochrane Library database	
Search number	Search query	Results	Search query	Results	Search query	Results
#1	methicillin-susceptible *Staphylococcus aureus*	3018	methicillin-susceptible *Staphylococcus aureus* [exp]	6835	methicillin-susceptible *Staphylococcus aureus*	176
#2	methicillin susceptible *Staphylococcus aureus*	5420	methicillin susceptible *Staphylococcus aureus* [exp]	6835	methicillin susceptible *Staphylococcus aureus*	238
#3	methicillin-sensitive *Staphylococcus aureus*	2300	methicillin-sensitive *Staphylococcus aureus* [exp]	6746	methicillin-sensitive *Staphylococcus aureus*	70
#4	methicillin sensitive *Staphylococcus aureus*	3920	methicillin sensitive *Staphylococcus aureus* [exp]	6746	methicillin sensitive *Staphylococcus aureus*	104
#5	MSSA	3257	MSSA	8884	MSSA	113
#6	#1 OR #2 OR #3 OR #4 OR #5	9488	#1 OR #2 OR #3 OR #4 OR #5	9786	#1 OR #2 OR #3 OR #4 OR #5	333
#7	bacteremia [mh]	45061	bacteremia [exp]	56699	bacteremia	2479
#8	bacteraemia	45061	bacteraemia [exp]	49825	bacteraemia	2479
#9	bloodstream infection	27505	bloodstream infection [exp]	15618	bloodstream infection	983
#10	#7 OR #8 OR #9	66994	#7 OR #8 OR #9	70992	#7 OR #8 OR #9	3218
#11	minimum inhibitory concentration	134806	minimum inhibitory concentration [exp]	107738	minimum inhibitory concentration	1152
#12	microbial sensitivity tests [mh]	128403	microbial sensitivity tests [mh]	9307	microbial sensitivity tests	1457
#13	MIC	39349	MIC	128645	MIC	1352
#14	#11 OR #12 OR #13	147130	#11 OR #12 OR #13	137364	#11 OR #12 OR #13	3219
#15	vancomycin [mh]	29888	vancomycin [mh]	95211	vancomycin	2008
#16	#6 AND #10 AND #14 AND #15	205	#6 AND #10 AND #14 AND #15	170	#6 AND #10 AND #14 AND #15	13

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Table S2 Raw data of mortality extracted from the included studies

Study	No. MSSA	No. death	High MIC		Low MIC		Odds ratios and 95% confidence intervals [a]		Mortality definition
			Events	Total	Events	Total	Unadjusted	Adjusted	
Abelenda 2018	58	15	6	23	10	35	**0.88 (0.27, 2.89)**	NA	30-day
Aguado 2011	99	14	6	23	8	76	**3.00 (0.92, 9.81)**	NA	30-day
Baxi 2016	230	37	11	39	26	191	2.47 (0.99, 8.96)	**1.71 (0.69, 5.35)** [d]	90-day
	230	27	7	39	20	191	1.95 (0.73, 6.75)	1.16 (0.42, 3.75) [d]	30-day
Bouiller 2018	250	64	18	64	46	186	**1.19 (0.63, 2.26)**	NA	30-day
Caston 2014	53	12	8	21	4	32	4.31 (1.10, 16.93)	**9.30 (1.31, 63.2)**	30-day
Cervera 2014	93	40	22	40	18	53	**2.38 (1.02, 5.52)**	NA	One-year
	93	37	21	40	16	53	2.56 (1.09, 6.00)	NA	In-hospital
Fernandez-Hidalgo 2018	173	58	19	60	39	113	0.88 (0.45, 1.72)	**0.87 (0.39, 1.92)**	In-hospital
Han 2012	202	25	NA	NA	NA	NA	1.47 (0.61, 3.54) [b]	**3.90 (1.07, 14.2)**	30-day
Holmes 2011	330	46	NA	NA	NA	NA	2.52 (1.33, 4.78) [b]	**2.44 (1.21, 4.92)**	30-day
Lopez-Cortes 2015	135	37	8	29	29	106	1.01 (0.40, 2.54)	**0.85 (0.32, 3.35)** [c]	30-day
	135	28	3	29	25	106	0.37 (0.10, 1.34)	0.33 (0.09, 1.50) [c]	14-day
San-Juan 2016	83	10	4	43	6	40	**0.58 (0.15, 2.23)**	NA	30-day
Song 2017	334	83	5	27	78	307	0.67 (0.24, 1.82)	NA	30-day (Etest)
	334	83	5	27	78	307	**1.54 (0.51, 4.66)**	NA	30-day (BMD)
Sullivan 2017	252	66	19	84	47	168	**0.75 (0.41, 1.39)**	NA	90-day
	252	59	18	84	41	168	0.84 (0.45, 1.58)	NA	60-day
	252	45	14	84	31	168	0.88 (0.44, 1.77)	NA	30-day
Videma 2014	84	32	25	55	7	29	**2. 62 (0.96, 7.14)**	NA	In-hospital
	84	17	13	55	4	29	1.93 (0.57, 6.59)	NA	30-day

[a] Data with bold font were used in the main analysis. Unadjusted odds ratios and 95% confidence intervals were calculated from the studies unless specified otherwise. Adjusted odds ratios and 95% confidence intervals were directly extracted from the studies unless specified otherwise.

[b] The unadjusted odds ratio and 95% confidence interval were directly extracted from the study.

[c] The adjusted odds ratio and 95% confidence interval were calculated from the study.

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Table S3 Raw data of septic thrombophlebitis extracted from the included studies

Study	No. MSSA	No. septic thrombophlebitis	High MIC		Low MIC		Odds ratios and 95% confidence intervals [a]	
			Events	Total	Events	Total	Unadjusted [b]	Adjusted
Aguado 2011	99	13	8	23	5	76	**7.57 (2.17, 26.39)**	NA
Bouiller 2018	250	15	3	64	12	186	**0.71 (0.19, 2.61)**	NA
Caston 2014	53	4	1	21	3	32	**0.48 (0.05, 4.99)**	NA
Cervera 2014	93	19	13	40	6	53	**3.77 (1.28, 11.07)**	NA
Lopez-Cortes 2015	135	2	2	29	0	106	**19.36 (0.90, 415,12)**	NA
San-Juan 2016	49	10	9	43	1	40	**10.32 (1.24, 85.71)**	NA

[a] Data with bold font were used in the main analysis.

[b] Unadjusted odds ratios and 95% confidence intervals were calculated from the studies

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Table S4 Raw data of persistent bacteraemia extracted from the included studies

Study	No. MSSA	No. persistent bacteraemia	High MIC		Low MIC		Odds ratios and 95% confidence intervals [a]	
			Events	Total	Events	Total	Unadjusted [b]	Adjusted
Abelenda 2018	58	16	9	23	7	35	**2.57 (0.79, 8.35)**	NA
Cervera 2014	93	9	4	40	5	53	**1.07 (0.27, 4.26)**	NA
Lopez-Cortes 2015	125	26	8	28	18	97	**1.76 (0.67, 4.62)**	NA
San-Juan 2016	83	6	4	43	2	40	**1.95 (0.34, 11.27)**	NA

[a] Data with bold font were used in the main analysis.

[b] Unadjusted odds ratios and 95% confidence intervals were calculated from the studies

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Table S5 Raw data of complicated bacteraemia extracted from the included studies

Study	No. MSSA	No. complicated bacteraemia	High MIC		Low MIC		Odds ratios and 95% confidence intervals [a]	
			Events	Total	Events	Total	Unadjusted	Adjusted
Aguado 2011	99	28	18	23	10	76	23.8 (7.20, 78.4)	**22.9 (6.7, 78.1)**
Bouiller 2018	250	83	10	64	73	186	**0.29 (0.14, 0.60)**	NA
Caston 2014	53	10	2	21	8	32	**0.32 (0.06, 1.66)**	NA
Falcon 2019	113	NA	NA	NA	NA	NA	**1.72 (0.66, 4.48)** [b] (BMD)	NA
	113	NA	NA	NA	NA	NA	1.69 (0.53, 5.38) [b] (Etest)	NA
Lopez-Cortes 2015	135	19	4	29	15	106	**0.97 (0.30, 3.19)**	NA
San-Juan 2016	83	26	18	43	8	40	**2.88 (1.08, 7.70)**	NA
Sullivan 2017	252	50	NA	NA	NA	NA	NA	**2.35 (1.26, 4.37)**

[a] Data with bold font were used in the main analysis. Unadjusted odds ratios and 95% confidence intervals were calculated from the studies unless specified otherwise. Adjusted odds ratios and 95% confidence intervals were directly extracted from the studies.

[b] The unadjusted odds ratio and 95% confidence intervals was directly extracted from the study

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Table S6 Assessment of the quality of the included studies

Study	Confounding	Selection of participants	Classification of interventions	Deviations from intended interventions	Missing data	Measurement of outcomes	Selection of the reported result	Overall
Abelenda 2018	Critical ROB	Serious ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Critical ROB
Aguado 2011	Moderate ROB	Moderate ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Moderate ROB
Baxi 2016	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB
Bouiller 2018	Critical ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Critical ROB
Caston 2014	Low ROB	Moderate ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Moderate ROB
Cervera 2014	Moderate ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Moderate ROB
Falcón1 2019	Critical ROB	Moderate ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Critical ROB
Fernandez-Hidalgo 2017	Moderate ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Moderate ROB
Han 2012	Low ROB	Moderate ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Moderate ROB
Holmes 2011	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Moderate ROB	Moderate ROB
Lopez-Cortes 2015	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB
San-Juan 2016	Critical ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Critical ROB
Song 2017	Critical ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Critical ROB
Sullivan 2017	Critical ROB	Moderate ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Critical ROB
Viedma 2014	Moderate ROB	Moderate ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Moderate ROB

Abbreviations: ROB, risk of bias.

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Table S7 Leave-one-out sensitivity analysis of mortality

Study omitted	OR (95% CI)	P Value	I^2 %
Abelenda 2018	1.47 (1.14-1.89)	0.003	43.1
Aguado 2011	1.39 (1.08-1.79)	0.01	40.7
Baxi 2016	1.42 (1.10-1.83)	0.007	44.6
Bouiller 2018	1.48 (1.14-1.94)	0.004	43.9
Caston 2014	1.39 (1.09-1.79)	0.009	33.9
Cervera 2014	1.37 (1.06-1.77)	0.016	40.8
Fernandez-Hidalgo 2018	1.52 (1.17-1.96)	0.002	40.3
Han 2012	1.38 (1.08-1.78)	0.011	38.2
Holmes 2011	1.33 (1.03-1.74)	0.032	37.8
Lopez-Cortes 2015	1.47 (1.14-1.89)	0.003	42.8
San-Juan 2016	1.48 (1.15-1.90)	0.002	40.0
Song 2017	1.43 (1.11-1.84)	0.005	44.9
Sullivan 2017	1.63 (1.25-2.13)	< 0.001	27.6
Viedma 2014	1.38 (1.07-1.78)	0.013	40.9

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Table S8 Definitions of the secondary outcomes

Study	Outcomes	Definitions of the secondary outcomes
Abelenda 2018 [13]	PB	PB was defined as positive blood cultures after more than 48 hours of appropriate antibiotic therapy.
Aguado. 2011 [8]	ST, CB	ST was defined by persistent bacteraemia at least 72 hours after initiation of active antimicrobial drugs and documented thrombi. CB was defined by one of the following events occurring after the first episode of bacteraemia: 1) development of endocarditis, septic thrombophlebitis, arthritis, spondylitis, as well as end-organ hematogenous spread of infection to other locations; 2) infection involving vascular or osteoarticular prostheses (excluding intravascular catheter) not removed within 4 days.
Bouiller 2018 [15]	ST, CB	The definition of ST was not mentioned. The definition of CB was the same as Aguado et al [8].
Caston 2014 [16]	ST, CB	The definitions of ST and CB were not mentioned.
Cervera 2014 [17]	ST, PB	The definition of ST was not mentioned. PB was defined as positive blood cultures ≥ 7 days after antimicrobial treatment.
Falcon 2019 [18]	CB	CB was defined by the following events: 1) endocarditis; 2) septic thrombophlebitis; 3) metastatic infection; 4) persistent bacteraemia for > 72 h after initiating active antimicrobial drug therapy, and 5) recurrent bacteraemia.
Lopez-Cortes 2015 [21]	ST, PB, CB	The definition of ST and CB were the same as Aguado et al [8]. PB was defined as the isolation of *S. aureus* in blood cultures obtained from peripheral veins for ≥ 3 days.
San-Juan 2016 [22]	ST, PB, CB	ST was defined by persistent bacteraemia for > 72 hours after the initiation of active therapy associated with the documentation of a thrombus at the site of catheter insertion. PB was defined as positive blood cultures after > 72 hours of appropriate antibiotic therapy. CB was defined by one of five events occurring after the incident episode of MSSA CRBSI: 1) development of endocarditis; 2) septic thrombophlebitis; 3) septic metastatic infection, including arthritis, spondylitis, infections involving vascular or osteoarticular prostheses (excluding intravascular catheter) or end-organ hematogenous seeding to other locations remote from the primary focus; 4) persistent bacteraemia for > 72 hours; or 5) persistent fever for > 4 days after extraction of the first blood culture yielding MSSA, provided that alternative causes had been reasonably excluded.
Sullivan 2017 [24]	CB	Complicated bacteraemia was defined by metastatic or embolic seeding, endovascular infection, or a duration > 3 days.

Abbreviations: CB, complicated bacteraemia; CRBSI, catheter-related bloodstream infections; MSSA, methicillin-susceptible *Staphylococcus aureus*; PB, persistent bacteraemia; SB, septic thrombophlebitis.

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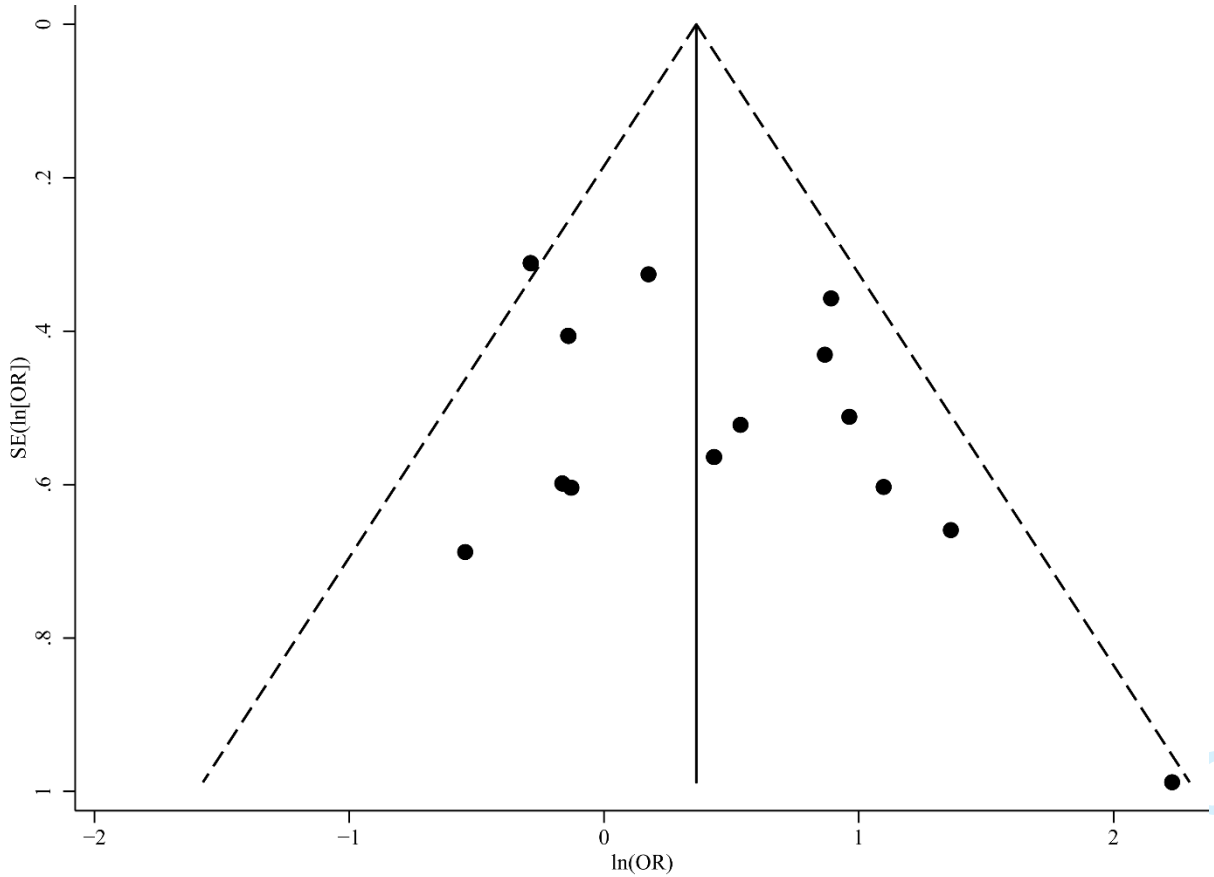


Figure S1 Funnel plot of mortality.



PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	2
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	6
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	6
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	6
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	6
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	6
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	7
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	7
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	10
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	10
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	11

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Effect of the vancomycin minimum inhibitory concentration on clinical outcomes in patients with methicillin-susceptible *Staphylococcus aureus* bacteraemia: a systematic review and meta-analysis

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ABSTRACT

Objective The use of the vancomycin minimum inhibitory concentration (MIC) as a prognostic predictor in patients with methicillin-susceptible *Staphylococcus aureus* (MSSA) has been debated in the last decade. We performed a systematic review and meta-analysis to investigate whether an elevated vancomycin MIC is associated with a worse prognosis for patients with MSSA bacteraemia.

Design Systematic review and meta-analysis

Data sources PubMed, EMBASE and the Cochrane Library were searched from inception to Dec 2019.

Eligibility criteria Randomized controlled trials or observational studies were considered eligible if they provided clinical outcomes of patients with MSSA bacteraemia, stratified by vancomycin MIC

Data synthesis Primary outcome was mortality. Secondary outcomes included septic thrombophlebitis, persistent bacteraemia and complicated bacteraemia. Pooled odds ratios (ORs) and 95% confidence intervals (CIs) were calculated. Subgroup analyses included susceptibility testing method.

Results Fifteen observational studies were included. Bacteraemia due to MSSA isolates with high vancomycin MICs was associated with higher mortality than isolates with low MICs (OR, 1.44; 95% CI, 1.12 to 1.84; I^2=40.3%). Additionally, significantly greater septic thrombophlebitis (OR, 3.16; 95% CI, 1.11 to 9.00; I^2=58.6%) and a trend towards more persistent bacteraemia (OR, 1.79; 95% CI, 0.97 to 3.31; I^2=0%) were observed in patients with high vancomycin MICs than in patients with low MICs. Differences in complicated bacteraemia were not significant. Similar findings were obtained in subgroup analyses using Etest. However, significant differences in outcomes were not observed between the high and low vancomycin MICs detected using broth microdilution.

Conclusions The available data suggest an association between elevated vancomycin MICs detected using Etest and adverse clinical outcomes for patients with MSSA bacteraemia. Future studies should validate these findings and explore the potential mechanisms.

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Strengths and limitations of this study

► This meta-analysis is the first study to comprehensively compare the clinical outcomes of patients with methicillin-susceptible *Staphylococcus aureus* bacteraemia with high and low vancomycin minimum inhibitory concentrations.

► Two authors independently performed the study selection, data extraction and quality assessment.

► Numerous sensitivity analyses were performed to examine the robustness of the findings.

► Subgroup analyses based on the susceptibility testing method were also performed.

► Due to insufficient details of empirical and definitive treatments, a subgroup analysis based on the antibiotic treatment could not be performed.

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INTRODUCTION

Staphylococcus aureus is one of the most common causes of healthcare-associated infections. Although more publications have focused on methicillin-resistant *S. aureus* (MRSA), bacteraemia caused by methicillin-susceptible *S. aureus* (MSSA) remains a significant global healthcare burden due to its high morbidity and mortality rates [1,2].

The recognition of potential risk factors is considered vital to improve the management of patients with MSSA bacteraemia. According to previous studies, β-lactams are superior to vancomycin in terms of survival for the treatment of MSSA bacteraemia [3-5]. Therefore, the current guidelines recommend switching therapy to a β-lactam antibiotic, such as cefazolin or anti-staphylococcal penicillins (ASPs), after the identification of an MSSA infection [6]. Furthermore, our previous meta-analysis [7] systematically evaluated studies focusing on the efficacy and safety of cefazolin versus ASPs in treating MSSA bacteraemia and found that cefazolin was significantly correlated with reductions in the death rate, clinical failure, hepatotoxicity and nephrotoxicity. The results favour cefazolin for the management of MSSA bacteraemia.

The choice of the antibiotic is a well-recognized risk factor for MSSA bacteraemia, but less is known about the effects of microbiological factors. A decade ago, two studies reported an association between a high vancomycin minimum inhibitory concentration (MIC) and a worse prognosis in terms of mortality and complicated bacteraemia among patients with MSSA bacteraemia, regardless of the antibiotic therapy used [8,9]. In 2014, Kalil et al. [10] conducted a meta-analysis to identify a potential relationship between vancomycin MICs and mortality for patients with *S. aureus* bloodstream infections. Unfortunately, a subgroup analysis of MSSA bacteraemia was unable to be performed because insufficient publications were available at that time. Since then, a growing number of studies have focused on this interesting issue but have reported inconsistent results. Therefore, we conducted a systematic review and meta-analysis to assess the effect of the vancomycin MIC on clinical outcomes in patients with MSSA bacteraemia.

METHODS

Literature search

PubMed, EMBASE and the Cochrane Library were searched from inception to Dec 2019 by two

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investigators. The search terms and strategies for this meta-analysis are summarized in Table S1. References from identified publications and relevant reviews were also checked to identify additional articles. No language restriction was imposed.

Inclusion criteria and study selection

Two authors independently screened the literature, and discrepancies were resolved through discussion. Randomized controlled trials or observational studies were considered eligible if they provided clinical outcomes of patients with MSSA bacteraemia, stratified by vancomycin MIC. Broth microdilution (BMD), automated BMD (e.g. Microscan®), and Etest were considered reasonable susceptibility testing methods. The following types of studies were excluded: (a) the study population did not consist of patients with MSSA bacteraemia, (b) solely paediatric studies, (c) no outcome data were available, (d) duplicate publications, and (e) conference abstracts.

Data extraction and quality assessment

Two reviewers independently extracted the data, and discrepancies were resolved by consensus. The following information was collected: study design, countries where the study was conducted, study period, sample size, patient characteristics (age, sex, comorbidity, severity of illness, and proportions of endocarditis), vancomycin MIC testing method, cut-off value, antimicrobial therapy used, and the clinical outcomes. The Risk of Bias in Nonrandomized Studies for Interventions (ROBINS-I) tool was used to assess the quality of observational studies [11].

Definitions and outcomes

The primary outcome was mortality. When data from more than one time point were available, the mortality at the latest point reporting the largest number of deaths was used in the main analysis (e.g., 30-day mortality had precedence over 14-day mortality). The secondary outcomes included septic thrombophlebitis, persistent bacteraemia, and complicated bacteraemia, as defined by the individual. If vancomycin MIC was detected using multiple methods, the data from the reference BMD method were adopted in the main analysis. We did not define specific cut-off points for the vancomycin MIC because different criteria were used in various studies.

Statistical analysis

The odds ratios (ORs) and 95% confidence intervals (CIs) for outcomes were calculated. If adjusted ORs were provided in the included studies, then they were used for data combination. Otherwise,

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unadjusted ORs were used for data combination. Heterogeneity was assessed using the I^2 statistic, and a value of > 50% was considered significant heterogeneity [12]. The fixed effects model was used when significant heterogeneity was not present; otherwise, the random effects model was used. For the outcomes using fixed effects model, we also calculated the random effects model for additional analyses to provide conservative pooling estimates. Subgroup analyses were performed based on the susceptibility testing method (Etest and BMD). Sensitivity analyses were conducted for the primary outcome by removing each study individually. Additional sensitivity analyses were performed by restricting studies that were classified according to a specific criterion, such as prospective, large-sample ($n \geq 90$), high-quality (low to moderate risk of bias in 'overall assessment' of ROBIN-I tool), unadjusted mortality rate, adjusted mortality rate, and 30-day mortality rate (the most frequent mortality rate reported). Publication bias was estimated by constructing a funnel plot and Egger's test. All the statistical analyses in the present meta-analysis were performed using STATA software version 15.0 (StataCorp, College Station, Texas, serial NO. 301506351850). The significance threshold was set to 0.05.

RESULTS

Search results

Fifteen studies [8,9,13-25] involving 2487 participants met the inclusion criteria. The details regarding identification of relevant studies are shown in Figure 1. Two studies [17,26] contained duplicate patients and the study with smaller sample size [26] was excluded.

All studies were published within the last decade, and the number of patients per study ranged from 53 to 334. Eight studies were prospective observational studies, and the others were retrospective studies. The proportion of females ranged from 28% to 54.7%, and the mean or median age ranged from 60 to 68 years. Most studies were conducted in Europe (n=10), followed by the USA (n=3) and South Korea (n=1); the remaining study was conducted in multiple countries (Australia and New Zealand). Most of the studies used the Etest method to determine the vancomycin MIC and chose an MIC value ≥ 1.5 μg/ml as the cut-off for a high vancomycin MIC. Additional details of the characteristics are presented in Table 1. Raw data for outcomes extracted from the included studies are presented in Tables S2-S5. Risk of bias assessment showed that six studies [13,15,18,22-24]

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had an overall critical risk of bias, and nine studies [8,9,14,16,17,19-21,25] had an overall low to moderate risk of bias (Table S6).

Primary outcome: mortality

Fourteen studies reported mortality rates [8,9,13-17,19-25]. The mortality rates of MSSA bacteraemia varied among studies from 12% to 43% (Table 1). Overall, the main analysis using a fixed effects model showed the mortality rate for patients infected with MSSA isolates with a high vancomycin MIC was significantly higher than isolates with a low MIC (OR, 1.44; 95% CI, 1.12 to 1.84; I^2=40.3%) (Figure 2). The additional analysis using the random effects model showed similar results (OR, 1.52; 95% CI, 1.08 to 2.12; I^2=40.3%) (Table 2). Sensitivity analyses limited to 14 studies providing unadjusted mortality rates (OR, 1.42; 95% CI, 1.13 to 1.79; I^2=36.0%), six studies providing adjusted mortality rates (OR, 1.74; 95% CI, 1.16 to 2.60; I^2=47.9%) and 12 studies reporting 30-day mortality rates (OR, 1.44; 95% CI, 1.09 to 1.92; I^2=28.2%) (Table 3) did not substantially alter the results of the main analysis. Similar results were obtained using the leave-one-out method (Table S7) and other prespecified sensitivity analyses (Table 3).

Secondary outcomes

Secondary outcome data were available in nine studies [8,13,15-18,21,22,24], and the definitions are shown in Table S8. Six studies reported septic thrombophlebitis [8,15-17,21,22]. An increased risk of septic thrombophlebitis was observed in patients with a high vancomycin MIC compared to patients with a low MIC in the pooled analysis using the random effects model (OR, 3.16; 95% CI, 1.11 to 9.00; I^2=58.6%) (Figure 3). Four studies reported persistent bacteraemia [13,17,21,22]. The meta-analysis using the fixed effects model revealed a non-significant higher risk of persistent bacteraemia in patients with a high vancomycin MIC (OR, 1.79; 95% CI, 0.97 to 3.31; I^2=0%) (Figure 4). The additional analysis using the random effects model supported the main conclusions (Table 2). Seven studies reported complicated bacteraemia [8,15,16,18,21,22,24], and no significant difference was observed in the pooled analysis using the random effects model (OR, 1.59; 95% CI, 0.58 to 4.37; I^2=87.0%) (Figure 5).

Subgroup analysis and publication bias

In the subset of studies using Etest, patients with a high vancomycin MIC still exhibited a significantly higher mortality rate (OR, 1.62; 95% CI, 1.21 to 2.17; I^2=27.6%) and septic

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thrombophlebitis (OR, 4.06; 95% CI, 1.40 to 11.80; I^2=57.3%) than patients with a low MIC. However, no significant differences in outcomes were observed between the high and low vancomycin MIC values detected using broth microdilution (BMD) (Table 2). The results of the additional analyses using the random effects model supported the main conclusions (Table 2). A visual inspection of the funnel plot showed symmetry (Figure S1), and the Egger test did not indicate significant publication bias (P=0.546).

DISCUSSION

To our knowledge, this systematic review and meta-analysis is the first to comprehensively compare the clinical outcomes of patients with MSSA bacteraemia presenting high and low vancomycin MICs. The main findings of the present study indicate that an elevated vancomycin MIC correlates with a higher mortality rate. The findings were relatively robust and were not significantly changed during the sensitivity analyses. In addition, patients with a high vancomycin MIC had a significantly higher risk of septic thrombophlebitis and a trend towards more persistent bacteraemia. No significant difference was observed in complicated bacteraemia. Based on these findings, the vancomycin MIC is potentially useful to predict adverse outcomes in patients with MSSA bacteraemia.

Several meta-analyses exploring the effect of the vancomycin MIC on the mortality rate in patients with a MRSA infection have produced positive correlations; the failure to achieve the pharmacokinetic/pharmacodynamic targets of vancomycin may be the main explanation [27-29]. However, vancomycin is not routinely used in clinical practice after the identification of an MSSA infection. In fact, the proportion of patients who received vancomycin for definitive therapy is less than one quarter in the included studies (Table 1). By analysing the subset of patients with MSSA bacteraemia treated with flucloxacillin, Holmes et al. observed a significant correlation between the 30-day mortality rate and a high vancomycin MIC [9]. Similar results were reported in the included study by Cervera et al., which included patients who exclusively received cloxacillin [17]. We hypothesise that a decreased vancomycin susceptibility might be a surrogate marker of a weak response to both vancomycin and other antibiotics (e.g., β-lactams) in the treatment of MSSA infections. Interestingly, MSSA isolates with decreased susceptibility to vancomycin also exhibit

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less susceptibility to the anti-staphylococcal penicillin and daptomycin [30]. This phenomenon might have contributed to special structural modifications in the cell walls of isolates with an elevated vancomycin MIC, particularly an increased cell wall thickness [31]. Moreover, reductions in autolysin activity and in the cell wall content of penicillin binding protein 4 may be involved in these changes [32].

The elevated vancomycin MIC might also be a marker for unfavourable bacterial or genetic factors that lead to adverse outcomes in patients with MSSA bacteraemia. The accessory gene regulator (*agr*) is a quorum-sensing regulator that is responsible for the expression of various virulence factors [33]. *Agr* dysfunction and *agr* type II have been correlated with an elevated vancomycin MIC, increased mortality rate, more persistent bacteraemia, and increased biofilm formation [25,34-37]. Moreover, we noticed that *agr* type III was less prevalent in MSSA isolates with a high vancomycin MIC than in isolates with a low MIC (17.2% vs. 34.0%) in the study by Lopez-Cortes et al. [21], although this difference did not reach statistical significance. In fact, the negative correlation between *agr* type III and mortality was observed in MSSA infective endocarditis [19].

Multiple clonal complexes (CCs), including CC5, CC8, CC88, and CC188, are associated with reduced vancomycin susceptibility in *S. aureus* bacteraemia isolates [25,34]. A recent study [38] analysed the genotypic profiles among 93 MSSA isolates reported in the study by Cervera et al. [17] and found that CC5 was significantly correlated with a higher mortality rate in the univariate analysis. Another included study [19] also revealed a correlation between MSSA isolates harbouring CC8 and increased in-hospital mortality in the univariate analysis.

Furthermore, specific resistance and virulence genes (including *blaZ, sea, clfA,* and *splA*) and the arginine catabolic mobile element locus were linked with an elevated vancomycin MIC in *S. aureus* bacteraemia [34]. MSSA isolates with genotypes *spa*-002 and *spa*-CC008 correlate with an elevated vancomycin MIC [24], and isolates with extracellular fibrinogen binding protein and serine endopeptidase are associated with a higher incidence of embolism [26].

Overall, the mechanism underlying the association between the increased vancomycin MIC and adverse outcomes in patients with MSSA bacteraemia appear to be complex. We postulate that this correlation is not attributed to a single cause, but results from the interactive effects of multiple factors. The findings of the present meta-analysis highlight the need for further studies to fill the

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knowledge gap about this interesting issue.

Notably, the most frequently used test to detect the vancomycin MIC in the included studies was the Etest. We identified a few studies performed susceptibility testing using Etest in parallel with manual or automated BMD methods. According to Han et al., the vancomycin MIC detected using Etest poorly correlated with the values obtained using the BMD method [20]. Song et al. reported the consistent underestimation of the MIC values from Etest among isolates with a high vancomycin MIC (2 mg/L) and overestimated among isolates with a low MIC (0.5 mg/L) using BMD [23]. As shown in the study by Sullivan et al., the vancomycin MIC values measured using automated BMD (Microscan®) are consistently higher than the values measured using Etest in the majority of MSSA isolates [24]. Because the application of the various susceptibility testing methods may introduce bias, we conducted a subgroup analysis according to the MIC methodology. In the subgroup analysis of studies using Etest, patients with a high vancomycin MIC still had a significantly higher mortality rate than patients with a low MIC. However, no significant difference was observed in the subgroup analysis of studies using BMD. Notably, we did not include the study by Holmes et al. [9] in the subgroup analysis of studies using BMD due to the unavailability of data, although the authors stated that a higher mortality rate was correlated with an increased vancomycin MIC detected using BMD. Therefore, the results of the subgroup analysis of studies using BMD may be underestimated. Nevertheless, the present meta-analysis indicated that Etest might be better for predicting clinical outcomes, although the reproducibility and interobserver variation may be a concern, as noted by Falcón et al. [39] Because the BMD is still the reference method for vancomycin susceptibility testing, more studies using the BMD are needed to validate the association between vancomycin MIC and clinical outcomes.

The present meta-analysis has several limitations that should be considered when interpreting the results. First, all studies included in the meta-analysis were observational studies containing many confounding factors. However, only six out of the included studies were designed to adjust for confounding factors and reported adjusted mortality rates. For studies without adjustment for potential confounders, the baseline characteristics of the patients (e.g., underlying conditions and the calculation of severity scores) between the high and low vancomycin MIC groups may influence the outcomes. For example, patients with high vancomycin MICs had numerically higher mean or

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median CCI scores in several studies [17,22]. Bouiller et al. observed more dialysis and a higher proportion of ultimately fatal disease in the high vancomycin MIC group [15]. More underlying diseases and higher CCI scores might be associated with adverse outcomes. Therefore, the adverse effect of the high vancomycin MIC on clinical outcomes might be overestimated. Second, almost half of the studies were retrospective in design, which may introduce inherent selection bias into this report. To address the potential confounding bias and selection bias, numerous sensitivity analyses were performed. Sensitivity analyses limited to prospective studies, high-quality studies based on the ROBIN-I tool, and studies providing adjusted mortality data did not substantially alter the results of the main analysis, further supporting the conclusions in our study. Third, as we mentioned in the INTRODUCTION section, the choice of antibiotic is closely related to the survival of patients with MSSA bacteraemia. However, the dose and duration of antibiotic therapy and the types of β-lactams chosen were not reported in most included studies. Due to insufficient details regarding both empirical and definitive treatments, a subgroup analysis based on the antibiotic treatment was unable to be conducted. Moreover, our previous findings regarding the differences in efficacy and safety between cefazolin and ASPs have not been sufficiently considered in the included studies [7]. Fourth, most MSSA isolates in the included studies had a vancomycin MIC ≤ 2 mg/L determined using BMD or ≤ 1.5 mg/L determined using Etest, and the outcomes of MSSA bacteraemia caused by more resistant isolates are unclear.

In conclusion, the available data suggest a correlation between a high vancomycin MIC detected using Etest and adverse clinical outcomes for patients with MSSA bacteraemia. More well-designed studies are warranted to validate the present findings and to explore the potential mechanisms.

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Contributors Nengming Lin contributed to conception and design; Changcheng Shi, Jian Ye, Renjie Xu, Weizhong Jin, Shuang Xu, Fei Teng, and Nengming Lin contributed to acquisition, analysis, and interpretation of data; Changcheng Shi were involved in drafting or critical revision of the manuscript. Changcheng Shi, Jian Ye, Renjie Xu, Weizhong Jin, Shuang Xu, Fei Teng, and Nengming Lin approved the final version.

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Patient consent for publication Not required

Data availability statement All data relevant to the study are included in the article or uploaded as supplementary information

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Table 1 Characteristics of the studies included in the meta-analysis

Study	Study design	Location	Study period	Sample size	Female %	Age, year	Assay method (HVM cut-off, mg/L)	Comorbidity & disease severity scores [a]	IE % [a]	Empirical treatment [a]	Definitive treatment [a]	Mortality %	Mortality definition
Abelenda 2018 [13]	Ret	Spain	2014-2016	58	41.4	68	BMD (≥2.0)	NR	5.2	Consistent with guidelines 29.31%, VAN 24.13%, DAP 1.72%	VAN 22.41%, DAP 5.17 %, LIN 6.89%, CLO 39.65%	25.9	30-day
Aguado. 2011 [8]	Ret	Spain	2002-2004	99	33.3	63	Etest (≥1.5)	Mean CCI 3.4 vs 2.8	7.1	GLY 78.3% vs 60.5%; ASBL 21.7% vs 26.3%	NR	14.1	30-day
Baxi 2016 [14]	Pro	USA	2008-2013	230	NR	NR	MicroScan® (≥2.0)	NR	NR	NR	NR	16.1 and 11.7	90-day, 30-day
Bouiller 2018 [15]	Pro	France	2009-2011	250	40.8	68	Etest (≥1.5)	NR	8.8	ASBL 57.8% vs 58.6%; VAN 9.4% vs 16.1%	Appropriate first line 96.9% vs 95.2%, optimal first line 15.6% vs 11.8%	25.6	30-day
Caston 2014 [16]	Ret	Spain	NR	53	54.7	68	MicroScan® (≥2.0)	Mean CCI 3.0 vs 2.9	5.7	ASBL 54.5% vs 40.6%; GLY 4.5% vs 19.4%; others 38.1% vs 34.3%	NR	22.6	30-day
Cervera 2014 [17]	Pro	Spain	1995-2011	93	28.0	60	Etest (≥1.5)	Median CCI 2 vs 1	100	NR	CLO 100% vs 100%	43.0 and 39.8	One-year and in-hospital
Falcon 2019 [18]	Ret	Spain	2013-2018	113	NR	NR	BMD (≥0.75 or 0.5), Etest (≥1.5)	NR	NR	NR	NR	NR	NR
Fernandez-Hidalgo 2018 [19]	Pro	Spain	2013-2016	173	39.1	65	Etest (≥1.5)	NR	100	ASBL 100% vs 100%	NR	33.5	In-hospital
Han 2012 [20]	Ret	USA	2007-2009	202	NR	NR	Etest (> 1.0)	NR	NR	VAN 60.9%	β-lactams 79.2%	12.4	30-day
Holmes 2011 [9]	Pro	Australia and New Zealand	2007-2008	330	32.4	60	Etest (> 1.5), BMD (NR)	NR	10.0	NR	FLU 80.6%, VAN 19.4%	13.9	30-day
Lopez-Cortes 2015 [21]	Pro	Spain	2008-2011	135	41.5	67	Etest (≥1.5)	Median CCI 1 vs 2, median PBS 1 vs 2	8.1	NR	CLO 76.9% vs 85.7%, GLY 0 vs 8.2%, ASBL other than CLO 15.4% vs 5.1%	27.4 and 20.7	30-day and 14-day
San-Juan 2016 [22]	Pro	Spain	2011-2014	83	41.0	60	Etest (≥1.5)	Mean CCI 4.0 vs 3.5, mean PBS 1.1 vs 1.5	2.4	ASBL 51.1% vs 52.5%, GLY 37.2% vs 40%, DAP 23.2% vs 17.5%	ASBL 100% vs 96.25%	12.0	30-day
Song 2017 [23]	Pro	South Korea	2009-2011	334	NR	NR	BMD (≥2.0), Etest (≥1.5)	NR	7.2	NR	Cephalosporin 48.1% vs 43.8%, NAF 29.6% vs 24.9%, GLY 22.2% vs 31.3%	24.9	30-day
Sullivan 2017 [24]	Ret	USA	2010-2012	252	41.3	60	MicroScan® (≥2.0), Etest (NR)	Mean CCI 5.1 vs 5.5, mean PBS 2.4 vs 2.2	NR	VAN 77% vs 62%; ASBL 17% vs 33%	NR	26.2, 23.4 and 17.8	90-day, 60-day, and 30-day
Viedma 2014 [25]	Ret	Spain	2010-2011	84	39.3	62	Etest (≥1.5)	NR	NR	NR	β-lactams 58.2% vs 65.5%, VAN 3.6% vs 24.1%	38.1 and 20.2	In-hospital and 30-day

Abbreviations: ASBL, anti-staphylococcal β-lactams; BMD, broth microdilution; CCI, Charlson Comorbidity Index; CLO, cloxacillin; DAP, daptomycin; FLU, flucloxacillin; GLY, glycopeptides; HVM, high vancomycin minimum inhibitory concentration; IE, infective endocarditis; LIN, linezolid; LVM, low vancomycin minimum inhibitory concentration; MIC, minimum inhibitory concentration; NAF, nafcillin; NR, not reported; PBS, Pitt bacteraemia score; Pro, prospective; Ret, retrospective; VAN, vancomycin.

[a] High vancomycin MIC vs low vancomycin MIC.

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Table 2 Subgroup analyses of outcomes

Outcomes	Susceptibility testing method	No. of studies	I^2 %	Estimates from fixed effect models		Estimates from random effect models	
				OR (95% CI)	P Value	OR (95% CI)	P Value
Mortality	Etest	10	27.6	1.62 (1.21-2.17)	0.001	1.63 (1.15, 2.32)	0.006
	BMD	5	45.7	1.12 (0.73-1.71)	0.617	1.32 (0.69, 2.52)	0.394
	Overall	14	40.3	1.44 (1.12-1.84)	0.004	1.52 (1.08-2.12)	0.015
Septic	Etest	5	57.3	Not performed	Not applicable	4.06 (1.40-11.8)	0.031
thrombophlebitis	BMD	1	Not available	Not performed	Not applicable	Not performed	Not applicable
	Overall	6	58.6	Not performed	Not applicable	3.16 (1.11-9.00)	0.031
Persistent	Etest	3	0	1.56 (0.76-3.21)	0.224	1.56 (0.76-3.21)	0.224
bacteraemia	BMD	1	Not available	Not performed	Not applicable	Not performed	Not applicable
	Overall	4	0	1.79 (0.97-3.31)	0.064	1.79 (0.97-3.31)	0.064
Complicated	Etest	5	90.1	Not performed	Not applicable	1.92 (0.45-8.25)	0.378
bacteraemia	BMD	3	59.0	Not performed	Not applicable	1.43 (0.58-3.51)	0.438
	Overall	7	87.0	Not performed	Not applicable	1.59 (0.58-4.37)	0.365

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Table 3 Sensitivity analyses of mortality

Term included	No. of studies included	OR (95% CI)	P Value	I^2 %
Prospective	8	1.42 (1.04-1.93)	0.026	13.0
Large-sample (n ≥ 90)	10	1.48 (1.04-2.11)	0.012	37.6
Low to moderate overall risk of bias	9	1.99 (1.43-2.76)	<0.001	27.0
Adjusted mortality	6	1.74 (1.16-2.60)	0.007	47.9
Unadjusted mortality	14	1.42 (1.13-1.79)	0.003	36.0
30-day mortality	12	1.44 (1.09-1.92)	0.011	28.2

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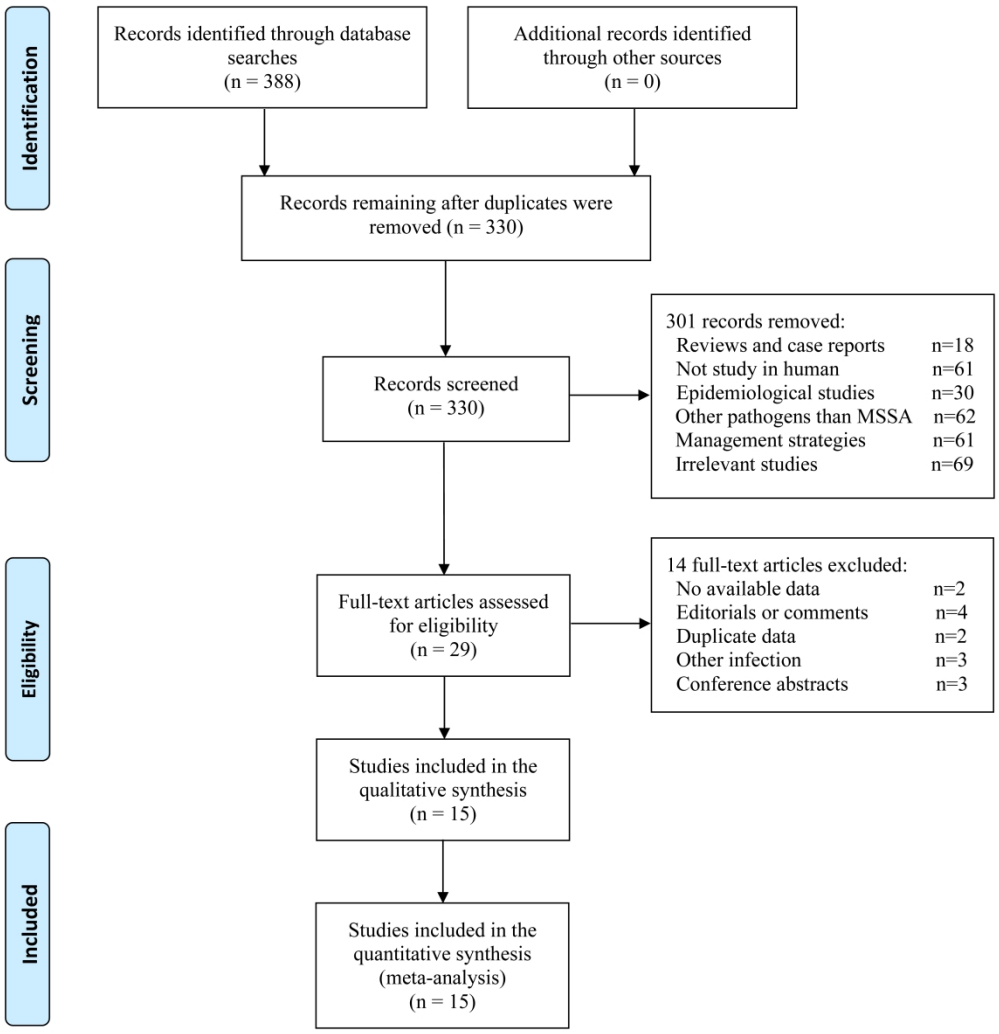


Figure 1 Flow diagram of the literature search and selection process.

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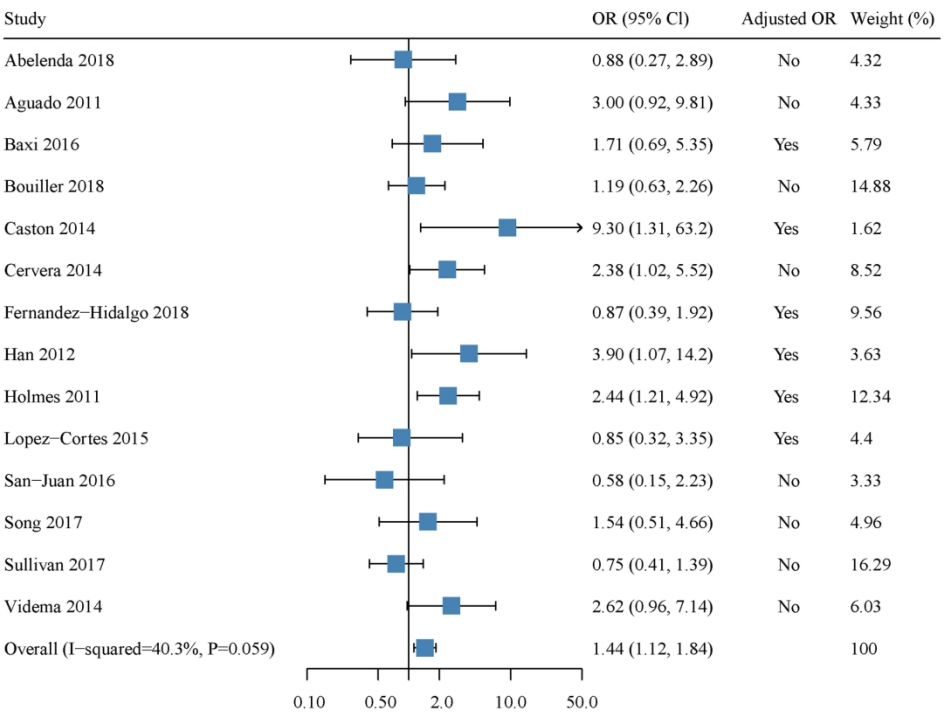


Figure 2 Forest plot of odds ratios for mortality.

169x123mm (300 x 300 DPI)

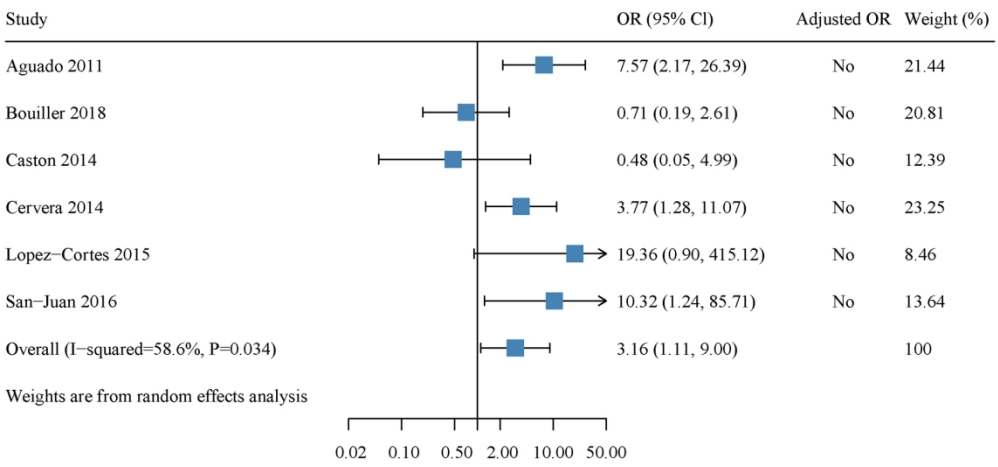


Figure 3 Forest plot of odds ratios for septic thrombophlebitis.

169x80mm (300 x 300 DPI)

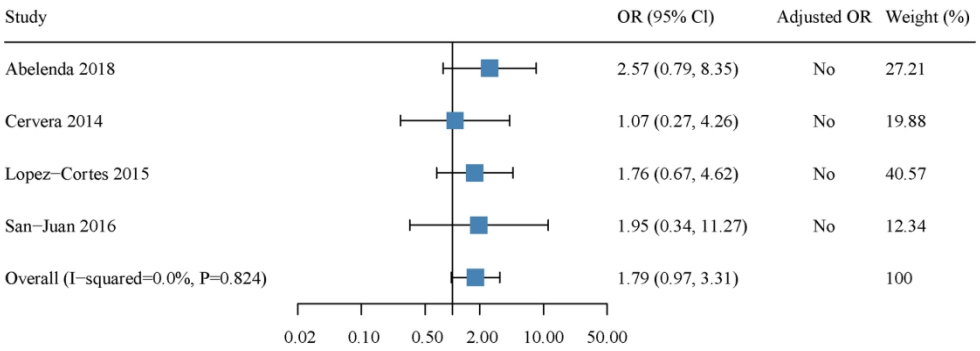


Figure 4 Forest plot of odds ratios for persistent bacteraemia.

169x61mm (300 x 300 DPI)

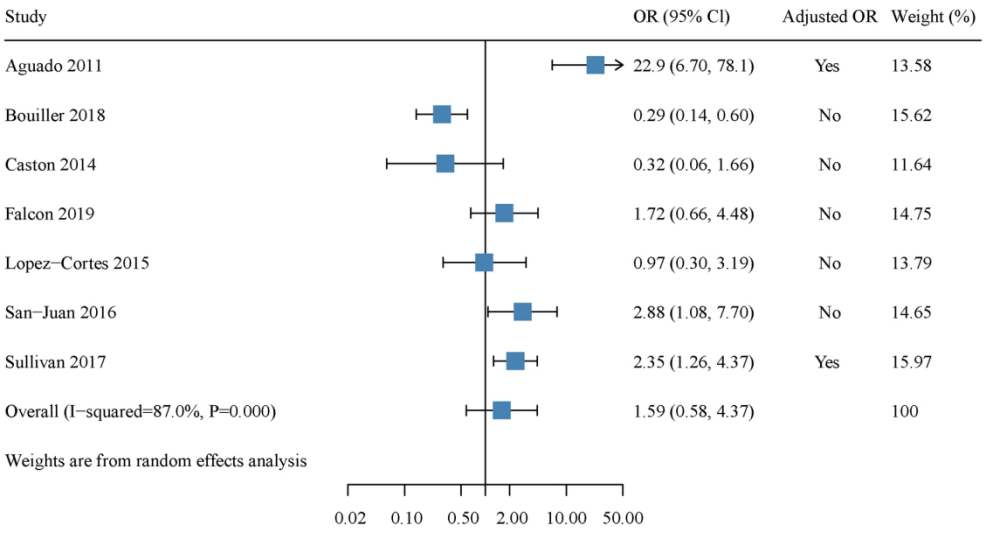


Figure 5 Forest plot of odds ratios for complicated bacteraemia.

169x92mm (300 x 300 DPI)

Table S1 Search strategy used in PubMed, EMBASE and the Cochrane Library databases

Database	PubMed		EMBASE		The Cochrane Library database	
Search number	Search query	Results	Search query	Results	Search query	Results
#1	methicillin-susceptible *Staphylococcus aureus*	3018	methicillin-susceptible *Staphylococcus aureus* [exp]	6835	methicillin-susceptible *Staphylococcus aureus*	176
#2	methicillin susceptible *Staphylococcus aureus*	5420	methicillin susceptible *Staphylococcus aureus* [exp]	6835	methicillin susceptible *Staphylococcus aureus*	238
#3	methicillin-sensitive *Staphylococcus aureus*	2300	methicillin-sensitive *Staphylococcus aureus* [exp]	6746	methicillin-sensitive *Staphylococcus aureus*	70
#4	methicillin sensitive *Staphylococcus aureus*	3920	methicillin sensitive *Staphylococcus aureus* [exp]	6746	methicillin sensitive *Staphylococcus aureus*	104
#5	MSSA	3257	MSSA	8884	MSSA	113
#6	#1 OR #2 OR #3 OR #4 OR #5	9488	#1 OR #2 OR #3 OR #4 OR #5	9786	#1 OR #2 OR #3 OR #4 OR #5	333
#7	bacteremia [mh]	45061	bacteremia [exp]	56699	bacteremia	2479
#8	bacteraemia	45061	bacteraemia [exp]	49825	bacteraemia	2479
#9	bloodstream infection	27505	bloodstream infection [exp]	15618	bloodstream infection	983
#10	#7 OR #8 OR #9	66994	#7 OR #8 OR #9	70992	#7 OR #8 OR #9	3218
#11	minimum inhibitory concentration	134806	minimum inhibitory concentration [exp]	107738	minimum inhibitory concentration	1152
#12	microbial sensitivity tests [mh]	128403	microbial sensitivity tests [mh]	9307	microbial sensitivity tests	1457
#13	MIC	39349	MIC	128645	MIC	1352
#14	#11 OR #12 OR #13	147130	#11 OR #12 OR #13	137364	#11 OR #12 OR #13	3219
#15	vancomycin [mh]	29888	vancomycin [mh]	95211	vancomycin	2008
#16	#6 AND #10 AND #14 AND #15	205	#6 AND #10 AND #14 AND #15	170	#6 AND #10 AND #14 AND #15	13

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Table S2 Raw data of mortality extracted from the included studies

Study	No. MSSA	No. death	High MIC		Low MIC		Odds ratios and 95% confidence intervals [a]		Mortality definition
			Events	Total	Events	Total	Unadjusted	Adjusted	
Abelenda 2018	58	15	6	23	10	35	**0.88 (0.27, 2.89)**	NA	30-day
Aguado 2011	99	14	6	23	8	76	**3.00 (0.92, 9.81)**	NA	30-day
Baxi 2016	230	37	11	39	26	191	2.47 (0.99, 8.96)	**1.71 (0.69, 5.35)** [d]	90-day
	230	27	7	39	20	191	1.95 (0.73, 6.75)	1.16 (0.42, 3.75) [d]	30-day
Bouiller 2018	250	64	18	64	46	186	**1.19 (0.63, 2.26)**	NA	30-day
Caston 2014	53	12	8	21	4	32	4.31 (1.10, 16.93)	**9.30 (1.31, 63.2)**	30-day
Cervera 2014	93	40	22	40	18	53	**2.38 (1.02, 5.52)**	NA	One-year
	93	37	21	40	16	53	2.56 (1.09, 6.00)	NA	In-hospital
Fernandez-Hidalgo 2018	173	58	19	60	39	113	0.88 (0.45, 1.72)	**0.87 (0.39, 1.92)**	In-hospital
Han 2012	202	25	NA	NA	NA	NA	1.47 (0.61, 3.54) [b]	**3.90 (1.07, 14.2)**	30-day
Holmes 2011	330	46	NA	NA	NA	NA	2.52 (1.33, 4.78) [b]	**2.44 (1.21, 4.92)**	30-day
Lopez-Cortes 2015	135	37	8	29	29	106	1.01 (0.40, 2.54)	**0.85 (0.32, 3.35)** [c]	30-day
	135	28	3	29	25	106	0.37 (0.10, 1.34)	0.33 (0.09, 1.50) [c]	14-day
San-Juan 2016	83	10	4	43	6	40	**0.58 (0.15, 2.23)**	NA	30-day
Song 2017	334	83	5	27	78	307	0.67 (0.24, 1.82)	NA	30-day (Etest)
	334	83	5	27	78	307	**1.54 (0.51, 4.66)**	NA	30-day (BMD)
Sullivan 2017	252	66	19	84	47	168	**0.75 (0.41, 1.39)**	NA	90-day
	252	59	18	84	41	168	0.84 (0.45, 1.58)	NA	60-day
	252	45	14	84	31	168	0.88 (0.44, 1.77)	NA	30-day
Videma 2014	84	32	25	55	7	29	**2. 62 (0.96, 7.14)**	NA	In-hospital
	84	17	13	55	4	29	1.93 (0.57, 6.59)	NA	30-day

[a] Data with bold font were used in the main analysis. Unadjusted odds ratios and 95% confidence intervals were calculated from the studies unless specified otherwise. Adjusted odds ratios and 95% confidence intervals were directly extracted from the studies unless specified otherwise.

[b] The unadjusted odds ratio and 95% confidence interval were directly extracted from the study.

[c] The adjusted odds ratio and 95% confidence interval were calculated from the study.

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Table S3 Raw data of septic thrombophlebitis extracted from the included studies

Study	No. MSSA	No. septic thrombophlebitis	High MIC		Low MIC		Odds ratios and 95% confidence intervals [a]	
			Events	Total	Events	Total	Unadjusted [b]	Adjusted
Aguado 2011	99	13	8	23	5	76	**7.57 (2.17, 26.39)**	NA
Bouiller 2018	250	15	3	64	12	186	**0.71 (0.19, 2.61)**	NA
Caston 2014	53	4	1	21	3	32	**0.48 (0.05, 4.99)**	NA
Cervera 2014	93	19	13	40	6	53	**3.77 (1.28, 11.07)**	NA
Lopez-Cortes 2015	135	2	2	29	0	106	**19.36 (0.90, 415,12)**	NA
San-Juan 2016	49	10	9	43	1	40	**10.32 (1.24, 85.71)**	NA

[a] Data with bold font were used in the main analysis.

[b] Unadjusted odds ratios and 95% confidence intervals were calculated from the studies

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Table S4 Raw data of persistent bacteraemia extracted from the included studies

Study	No. MSSA	No. persistent bacteraemia	High MIC		Low MIC		Odds ratios and 95% confidence intervals [a]	
			Events	Total	Events	Total	Unadjusted [b]	Adjusted
Abelenda 2018	58	16	9	23	7	35	**2.57 (0.79, 8.35)**	NA
Cervera 2014	93	9	4	40	5	53	**1.07 (0.27, 4.26)**	NA
Lopez-Cortes 2015	125	26	8	28	18	97	**1.76 (0.67, 4.62)**	NA
San-Juan 2016	83	6	4	43	2	40	**1.95 (0.34, 11.27)**	NA

[a] Data with bold font were used in the main analysis.

[b] Unadjusted odds ratios and 95% confidence intervals were calculated from the studies

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Table S5 Raw data of complicated bacteraemia extracted from the included studies

Study	No. MSSA	No. complicated bacteraemia	High MIC		Low MIC		Odds ratios and 95% confidence intervals [a]	
			Events	Total	Events	Total	Unadjusted	Adjusted
Aguado 2011	99	28	18	23	10	76	23.8 (7.20, 78.4)	**22.9 (6.7, 78.1)**
Bouiller 2018	250	83	10	64	73	186	**0.29 (0.14, 0.60)**	NA
Caston 2014	53	10	2	21	8	32	**0.32 (0.06, 1.66)**	NA
Falcon 2019	113	NA	NA	NA	NA	NA	**1.72 (0.66, 4.48)** [b] (BMD)	NA
	113	NA	NA	NA	NA	NA	1.69 (0.53, 5.38) [b] (Etest)	NA
Lopez-Cortes 2015	135	19	4	29	15	106	**0.97 (0.30, 3.19)**	NA
San-Juan 2016	83	26	18	43	8	40	**2.88 (1.08, 7.70)**	NA
Sullivan 2017	252	50	NA	NA	NA	NA	NA	**2.35 (1.26, 4.37)**

[a] Data with bold font were used in the main analysis. Unadjusted odds ratios and 95% confidence intervals were calculated from the studies unless specified otherwise. Adjusted odds ratios and 95% confidence intervals were directly extracted from the studies.

[b] The unadjusted odds ratio and 95% confidence intervals was directly extracted from the study

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Table S6 Assessment of the quality of the included studies

Study	Confounding	Selection of participants	Classification of interventions	Deviations from intended interventions	Missing data	Measurement of outcomes	Selection of the reported result	Overall
Abelenda 2018	Critical ROB	Serious ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Critical ROB
Aguado 2011	Moderate ROB	Moderate ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Moderate ROB
Baxi 2016	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB
Bouiller 2018	Critical ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Critical ROB
Caston 2014	Low ROB	Moderate ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Moderate ROB
Cervera 2014	Moderate ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Moderate ROB
Falcón1 2019	Critical ROB	Moderate ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Critical ROB
Fernandez-Hidalgo 2017	Moderate ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Moderate ROB
Han 2012	Low ROB	Moderate ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Moderate ROB
Holmes 2011	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Moderate ROB	Moderate ROB
Lopez-Cortes 2015	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB
San-Juan 2016	Critical ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Critical ROB
Song 2017	Critical ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Critical ROB
Sullivan 2017	Critical ROB	Moderate ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Critical ROB
Viedma 2014	Moderate ROB	Moderate ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Moderate ROB

Abbreviations: ROB, risk of bias.

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Table S7 Leave-one-out sensitivity analysis of mortality

Study omitted	OR (95% CI)	P Value	I^2 %
Abelenda 2018	1.47 (1.14-1.89)	0.003	43.1
Aguado 2011	1.39 (1.08-1.79)	0.01	40.7
Baxi 2016	1.42 (1.10-1.83)	0.007	44.6
Bouiller 2018	1.48 (1.14-1.94)	0.004	43.9
Caston 2014	1.39 (1.09-1.79)	0.009	33.9
Cervera 2014	1.37 (1.06-1.77)	0.016	40.8
Fernandez-Hidalgo 2018	1.52 (1.17-1.96)	0.002	40.3
Han 2012	1.38 (1.08-1.78)	0.011	38.2
Holmes 2011	1.33 (1.03-1.74)	0.032	37.8
Lopez-Cortes 2015	1.47 (1.14-1.89)	0.003	42.8
San-Juan 2016	1.48 (1.15-1.90)	0.002	40.0
Song 2017	1.43 (1.11-1.84)	0.005	44.9
Sullivan 2017	1.63 (1.25-2.13)	< 0.001	27.6
Viedma 2014	1.38 (1.07-1.78)	0.013	40.9

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Table S8 Definitions of the secondary outcomes

Study	Outcomes	Definitions of the secondary outcomes
Abelenda 2018 [13]	PB	PB was defined as positive blood cultures after more than 48 hours of appropriate antibiotic therapy.
Aguado. 2011 [8]	ST, CB	ST was defined by persistent bacteraemia at least 72 hours after initiation of active antimicrobial drugs and documented thrombi. CB was defined by one of the following events occurring after the first episode of bacteraemia: 1) development of endocarditis, septic thrombophlebitis, arthritis, spondylitis, as well as end-organ hematogenous spread of infection to other locations; 2) infection involving vascular or osteoarticular prostheses (excluding intravascular catheter) not removed within 4 days.
Bouiller 2018 [15]	ST, CB	The definition of ST was not mentioned. The definition of CB was the same as Aguado et al [8].
Caston 2014 [16]	ST, CB	The definitions of ST and CB were not mentioned.
Cervera 2014 [17]	ST, PB	The definition of ST was not mentioned. PB was defined as positive blood cultures ≥ 7 days after antimicrobial treatment.
Falcon 2019 [18]	CB	CB was defined by the following events: 1) endocarditis; 2) septic thrombophlebitis; 3) metastatic infection; 4) persistent bacteraemia for > 72 h after initiating active antimicrobial drug therapy, and 5) recurrent bacteraemia.
Lopez-Cortes 2015 [21]	ST, PB, CB	The definition of ST and CB were the same as Aguado et al [8]. PB was defined as the isolation of *S. aureus* in blood cultures obtained from peripheral veins for ≥ 3 days.
San-Juan 2016 [22]	ST, PB, CB	ST was defined by persistent bacteraemia for > 72 hours after the initiation of active therapy associated with the documentation of a thrombus at the site of catheter insertion. PB was defined as positive blood cultures after > 72 hours of appropriate antibiotic therapy. CB was defined by one of five events occurring after the incident episode of MSSA CRBSI: 1) development of endocarditis; 2) septic thrombophlebitis; 3) septic metastatic infection, including arthritis, spondylitis, infections involving vascular or osteoarticular prostheses (excluding intravascular catheter) or end-organ hematogenous seeding to other locations remote from the primary focus; 4) persistent bacteraemia for > 72 hours; or 5) persistent fever for > 4 days after extraction of the first blood culture yielding MSSA, provided that alternative causes had been reasonably excluded.
Sullivan 2017 [24]	CB	Complicated bacteraemia was defined by metastatic or embolic seeding, endovascular infection, or a duration > 3 days.

Abbreviations: CB, complicated bacteraemia; CRBSI, catheter-related bloodstream infections; MSSA, methicillin-susceptible *Staphylococcus aureus*; PB, persistent bacteraemia; SB, septic thrombophlebitis.

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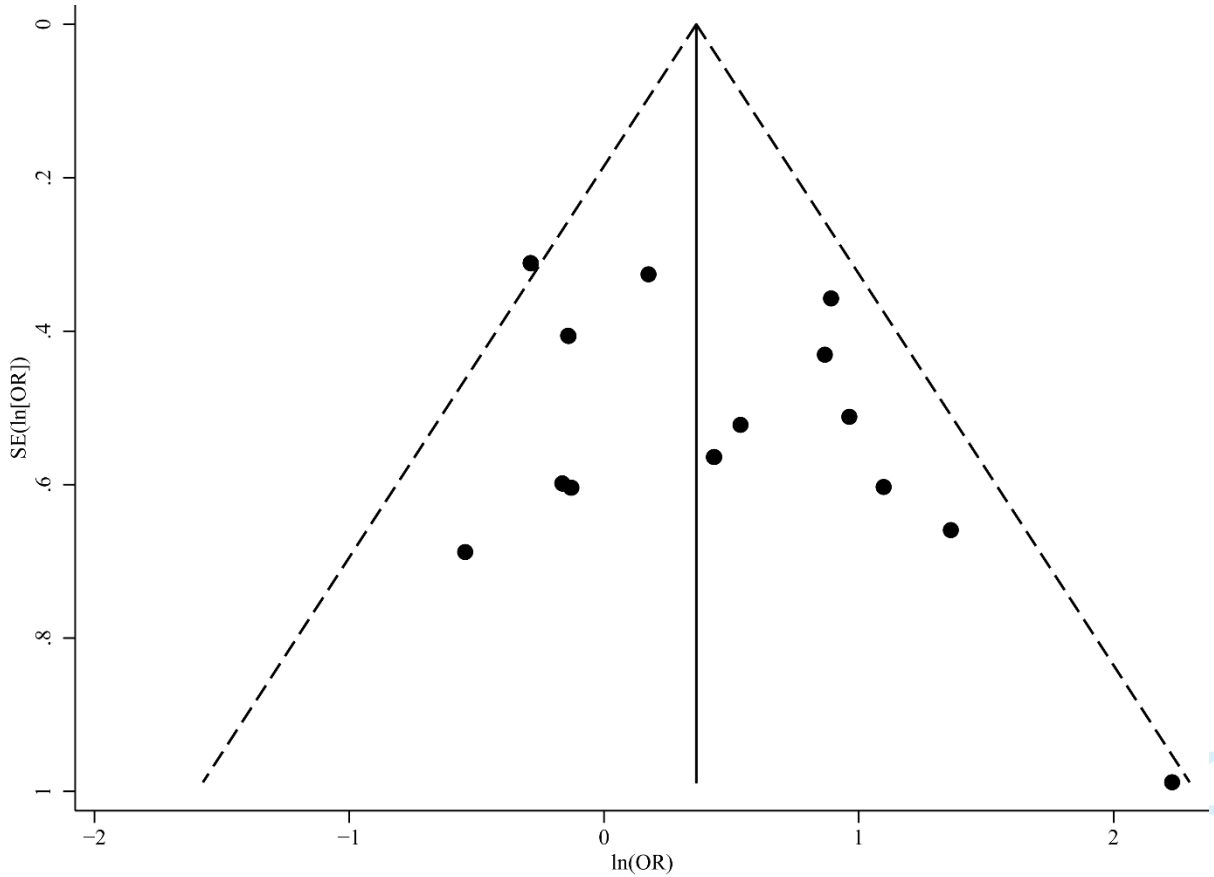


Figure S1 Funnel plot of mortality.



PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	2
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	6
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	6
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	6
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	6
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	6
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	7
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	7
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	10
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	10
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	11

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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